

# MEMORIA TÉCNICA

## Xiaomi ElectricScooter4

Ingeniería Inversa y Delineación de Planos

### Introducción

Este documento compila el levantamiento gráfico integral del vehículo **Xiaomi Electric Scooter 4**. El trabajo se rige por la normativa **UNE** para la representación de vistas y acotación funcional.

Para la organización corrección y distribución de planos se ha utilizado una pagina web creada específicamente para este proyecto. Puedes acceder aquí abajo.

Dashboard del Proyecto PEL →

### Estructura Organizativa del Proyecto

#### DIRECCIÓN Y JEFATURAS

- **Director del Proyecto:** Eloy Santos de Tourdonnet
- **Jefes de PL:** Marcos Menéndez (PL 19), Eloy Santos (PL 20), Hugo Ojeda (PL 21)

#### INGENIEROS TÉCNICOS ASIGNADOS

\*Nombres en rojo indican bajas comunicadas.

##### Equipo PL 19

- Barbu, Claudiu R.
- Fernández R., Darío
- Fuentes Díaz, Daniel
- Huebra Garnero, J.
- Núñez Bareiro, N.
- Quintana B., Paula
- Rodríguez O., M.
- Secades G., Daniel Silva
- Ridao, Jacobo

##### Equipo PL 20

- Aguilar L., S.
- Alonso Alonso, Julio
- Álvarez Estévez, Alejandro
- Berjano Ces, Iyán
- Birca, Carlos Alberto
- Fraile F., Pablo
- Iordache, Rubén
- Osagiede M., Lola Rivas
- Piñeiro, Pablo
- Sánchez C., Lorenzo

##### Equipo PL 21

- Dou González, Iván
- Fernández Uña, V.
- Menéndez G., Marta
- Mohamed Lagdof A.
- Moreiras M., Ian
- Servera E., Mark
- Villanueva F., F.

### Directorio de Planos

#### 1. Piezas Individuales

Dirección: 0.1

PL 19: 1.1 / 1.2 / 1.3 / 1.4 / 1.5 / 1.6 / 1.7 / 1.8 / 1.9 / 1.10(I) / 1.10(II)

PL 20: 2.1 / 2.2(I) / 2.2(II) / 2.3(I) / 2.3(II) / 2.4 / 2.5 / 2.6 / 2.7 / 2.8 / 2.9 / 2.10 / 2.11

PL 21: 3.1(I) / 3.1(II) / 3.2(I) / 3.2(II) / 3.3 / 3.4 / 3.5 / 3.6 / 3.7

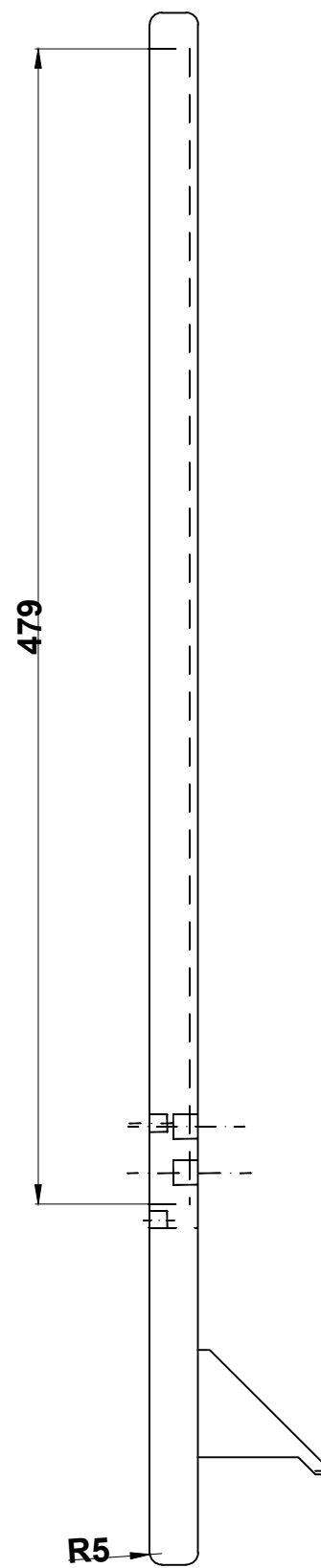
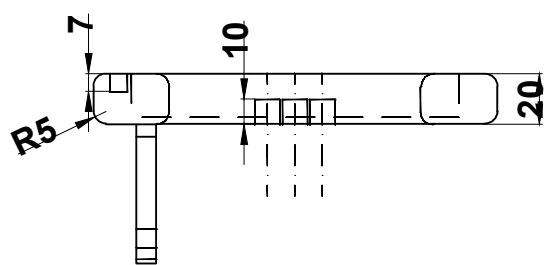
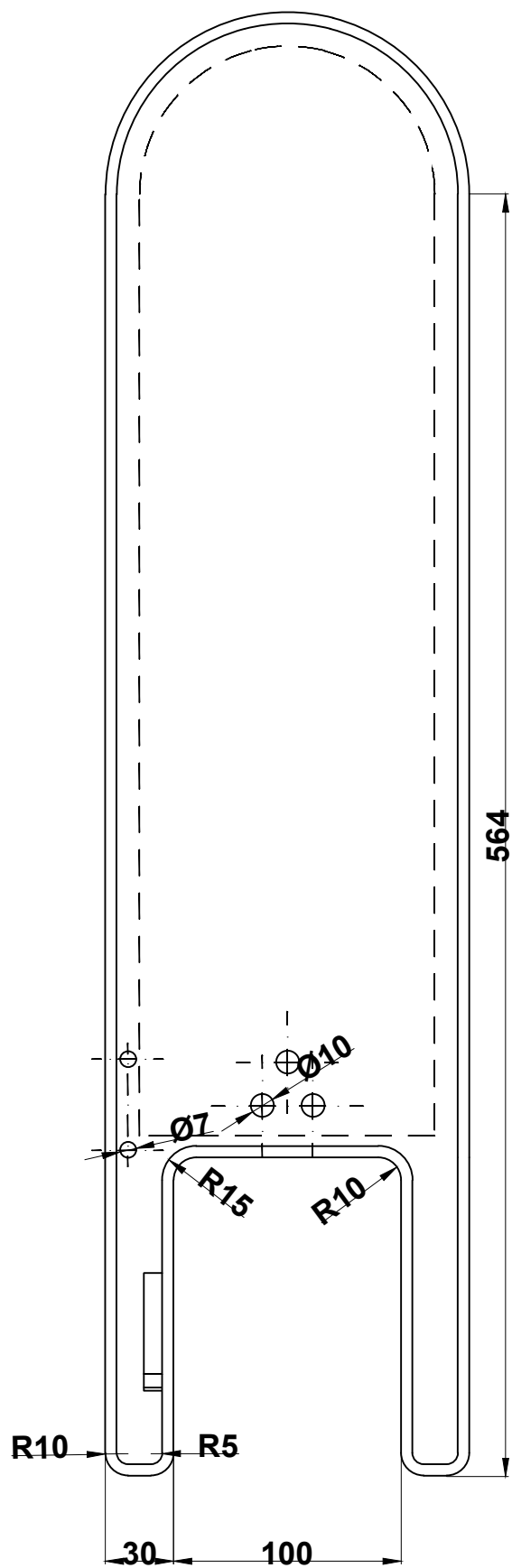
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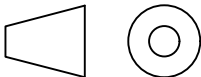
S1 / S2 / S3 / S4 / S5 / S6

#### 3. Proyecto Completo

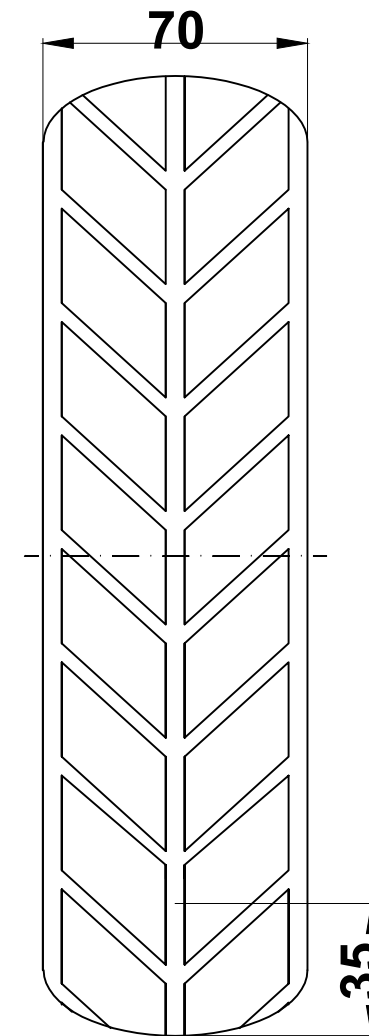
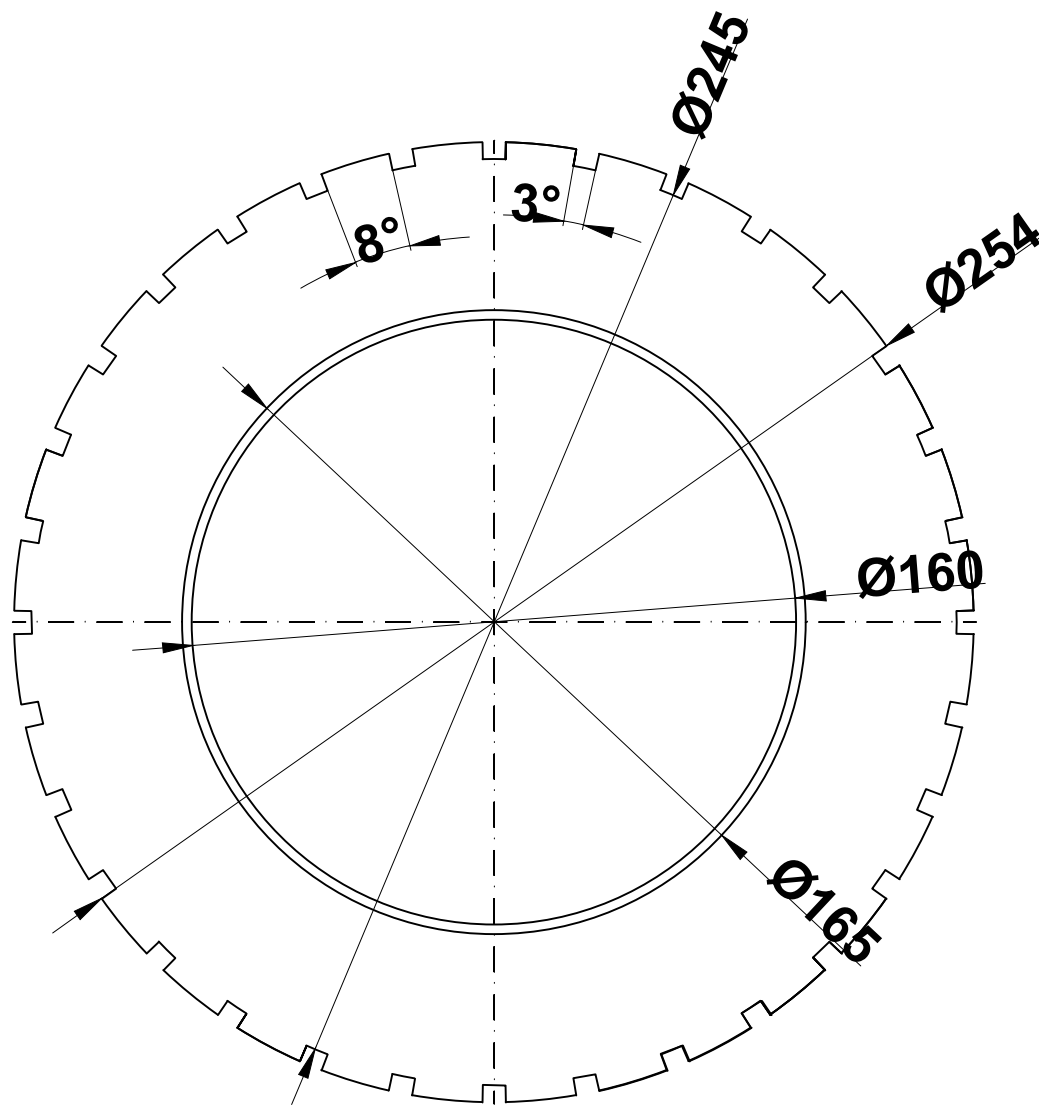
Ensamblaje General

Gijón, abril de 2026

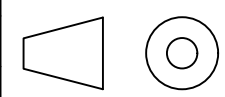


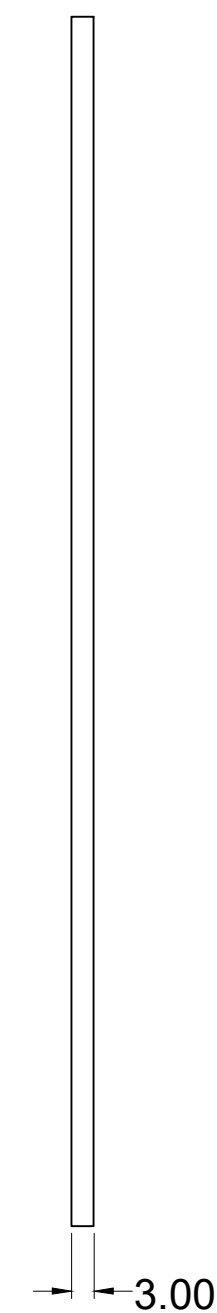
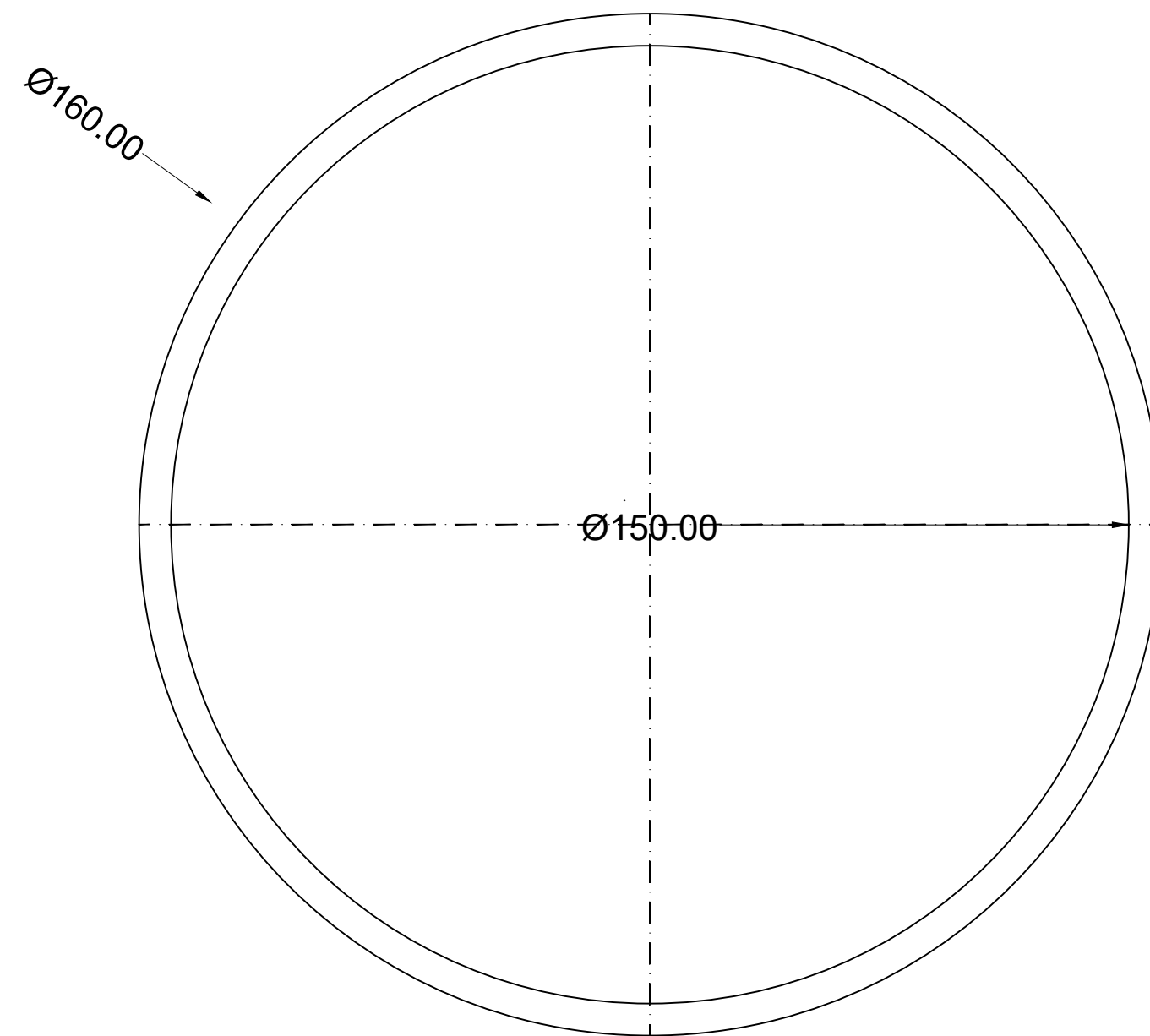
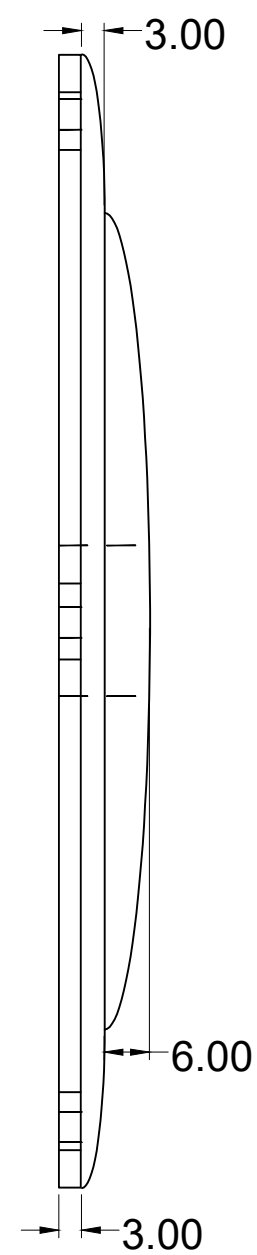
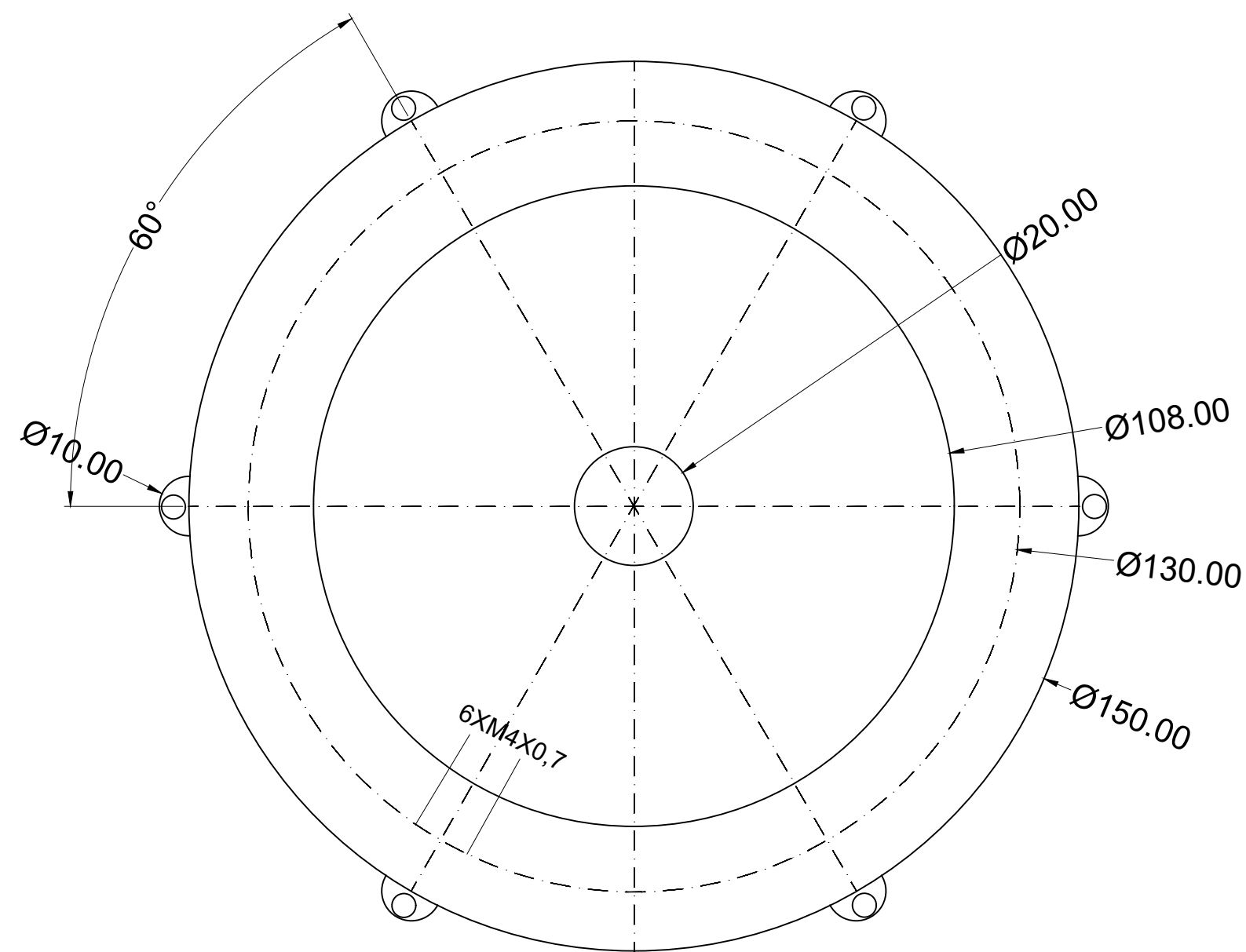
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| GRUPO PA                                           | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                              | PL 20            | 0.1               | MM                                                                                    |
| PLATAFORMA BASE                                    |                  |                   | ESCALA 1:4                                                                            |
| INGENIERO/A RESPONSABLE: Eloy Santos de Tourdonnet |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                           | ACERO            | UNE 1-032-82      |                                                                                       |



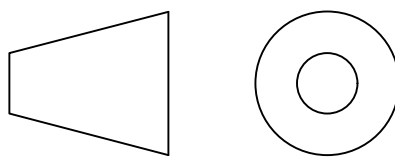


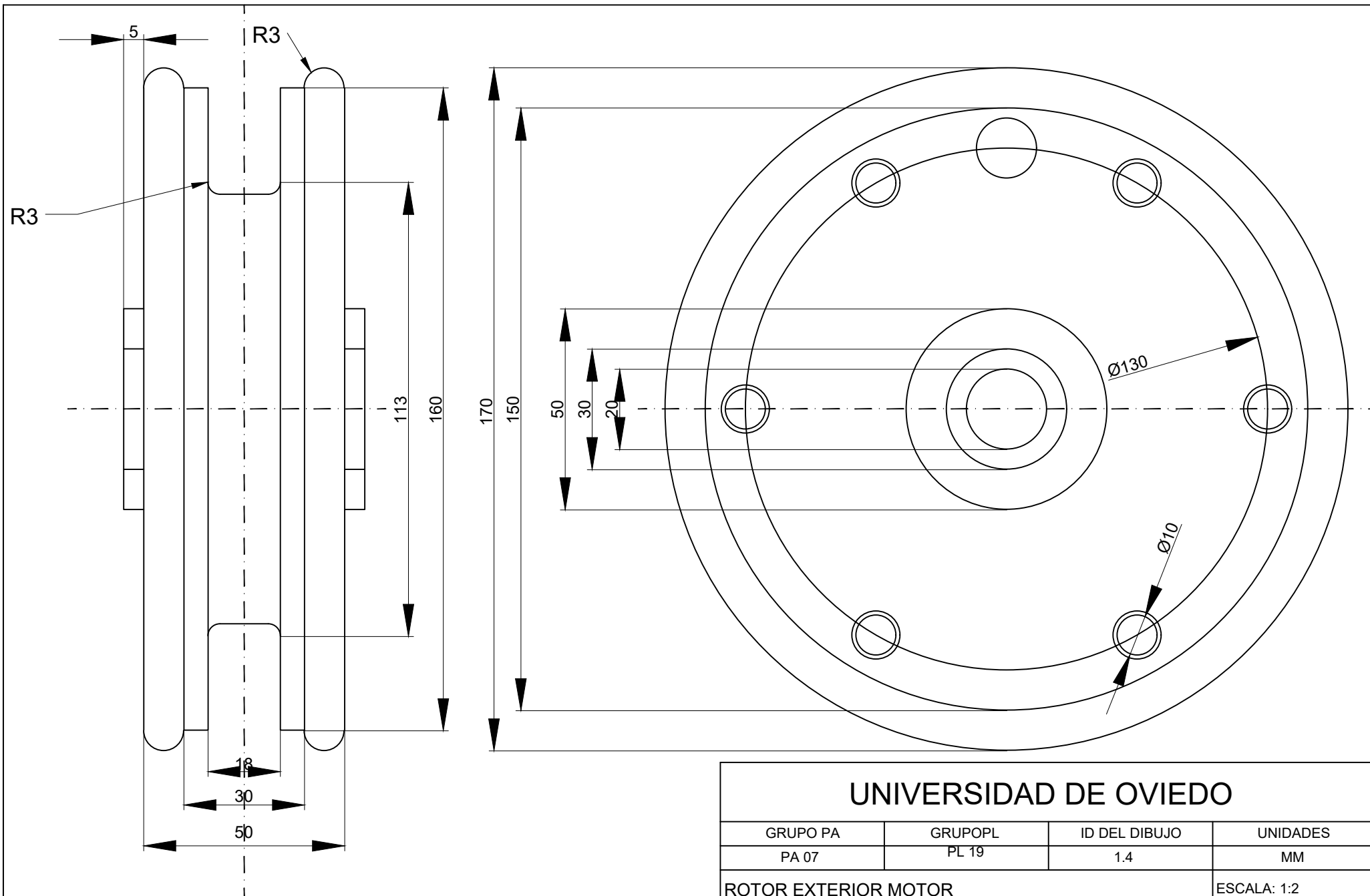
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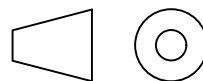
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| PA 07                                  | PL 19            | 1.2               | MM                                                                                    |
| Neumáticos 10"                         |                  |                   | ESCALA 1:4                                                                            |
| INGENIERO/A RESPONSABLE: Javier Huebra |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez         |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                     | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| Caucho                                 | Goma             | UNE 1-032-82      |                                                                                       |

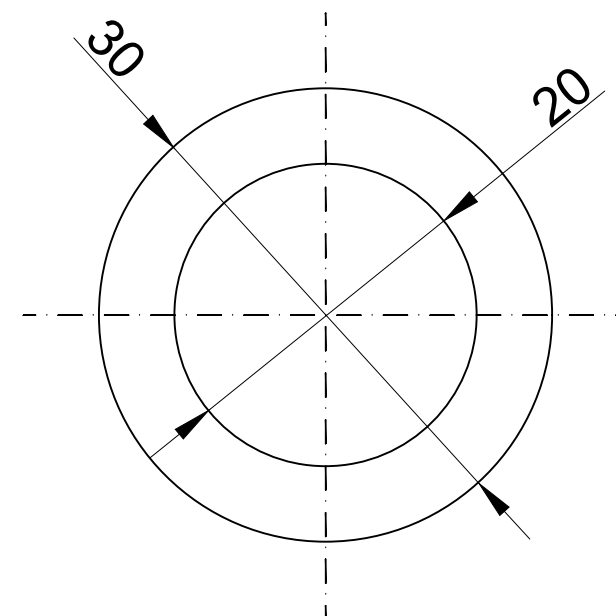
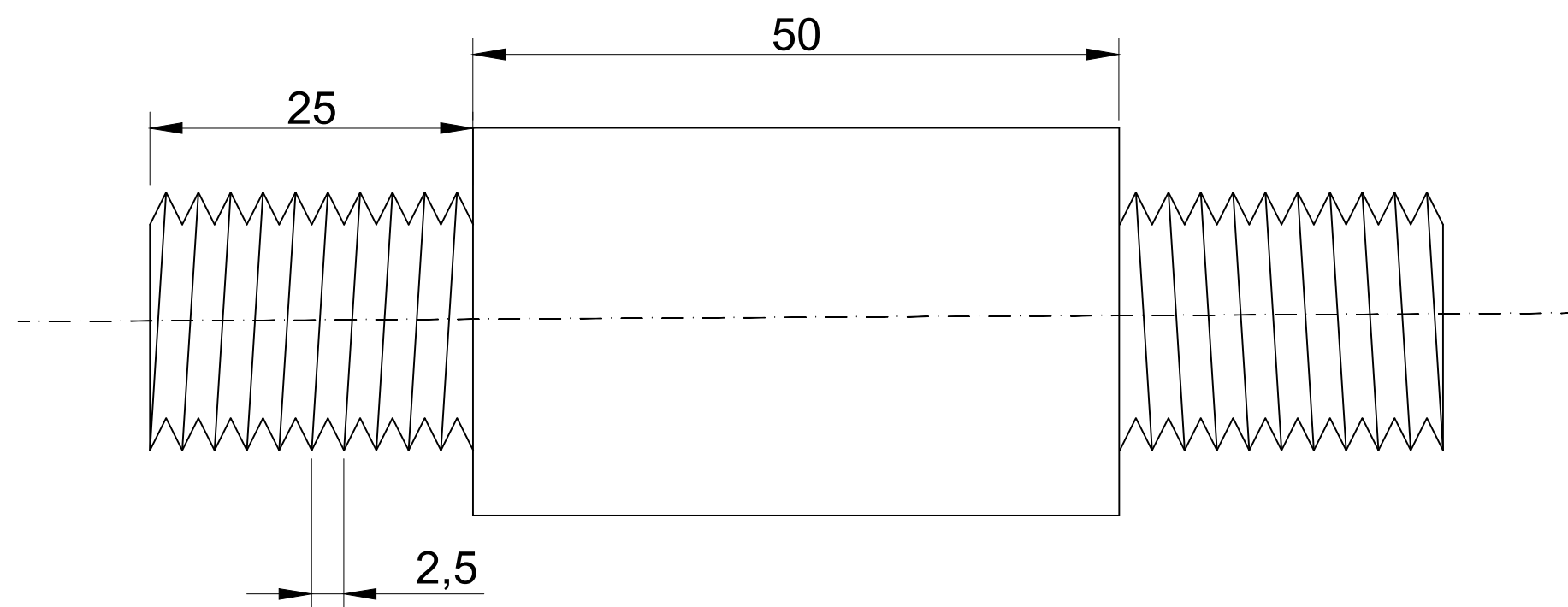
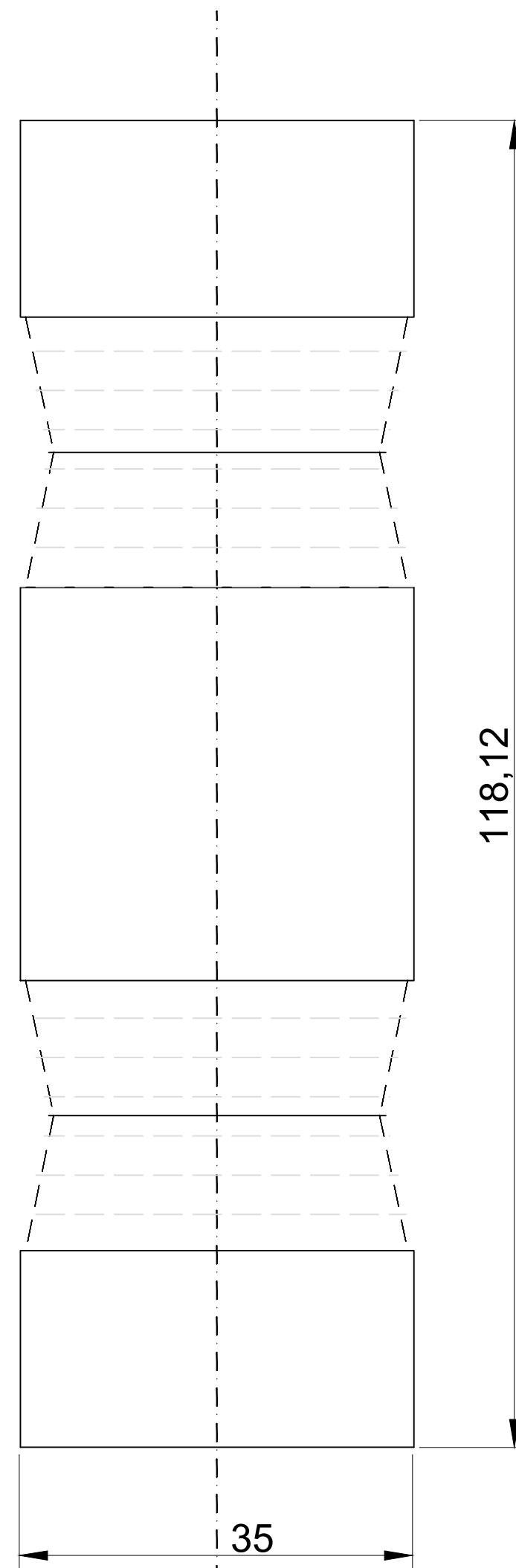
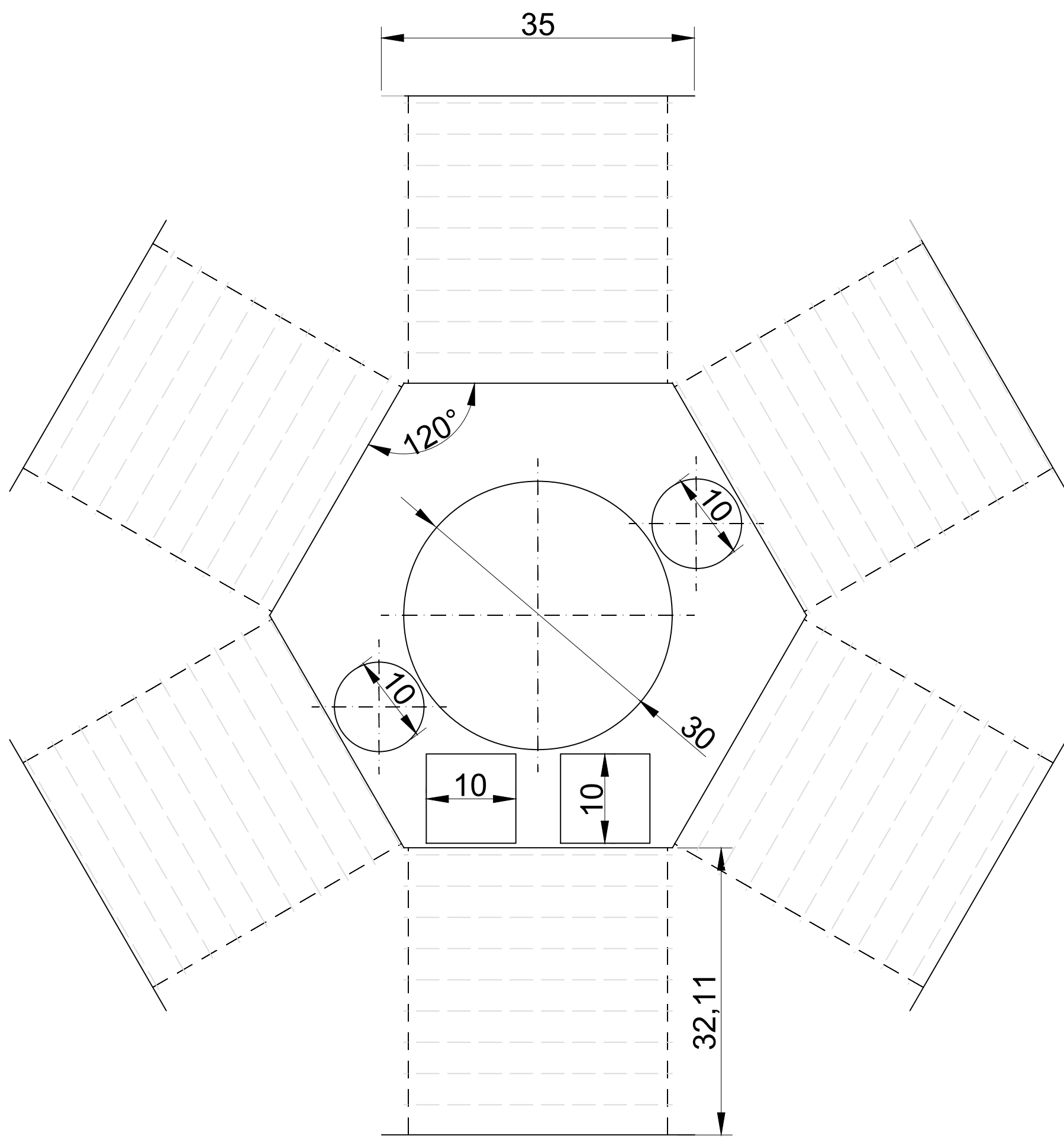


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| GRUPO PA                                      | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                         | PL 19            | 1.3               | MM                                                                                    |
| TAPAS DEL MOTOR                               |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Nicole Núñez Bareiro |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez                |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                            | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                      | PLÁSTICO         | UNE 1-032-82      |                                                                                       |

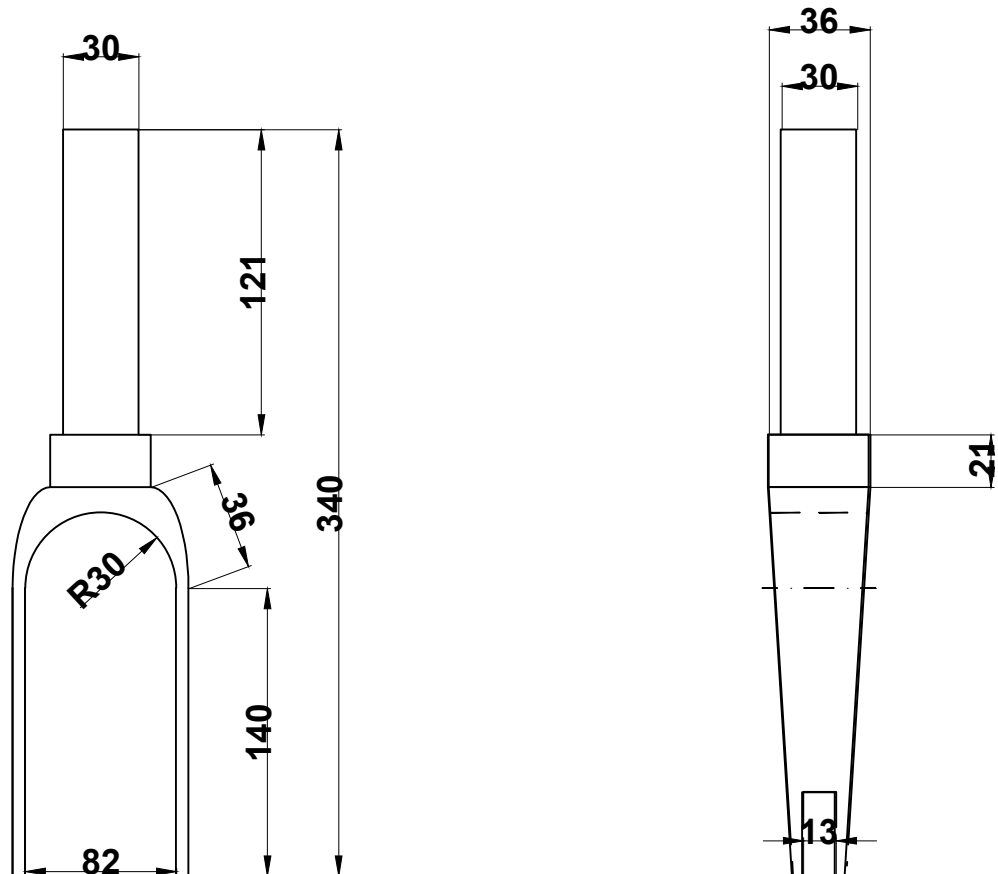


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|-----------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                   | PL 19            | 1.4               | MM                                                                                    |
| ROTOR EXTERIOR MOTOR                    |                  |                   | ESCALA: 1:2                                                                           |
| INGENIERO/A RESPONSABLE: Daniel Secades |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez          |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                      | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ACERO                                   | PLÁSTICO         | UNE 1-032-82      |                                                                                       |



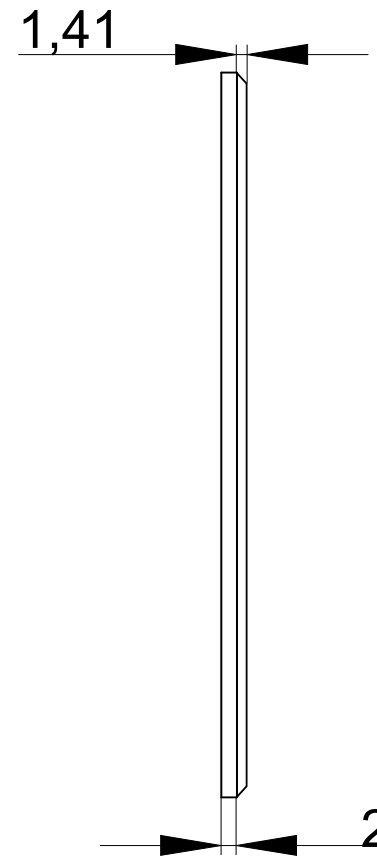
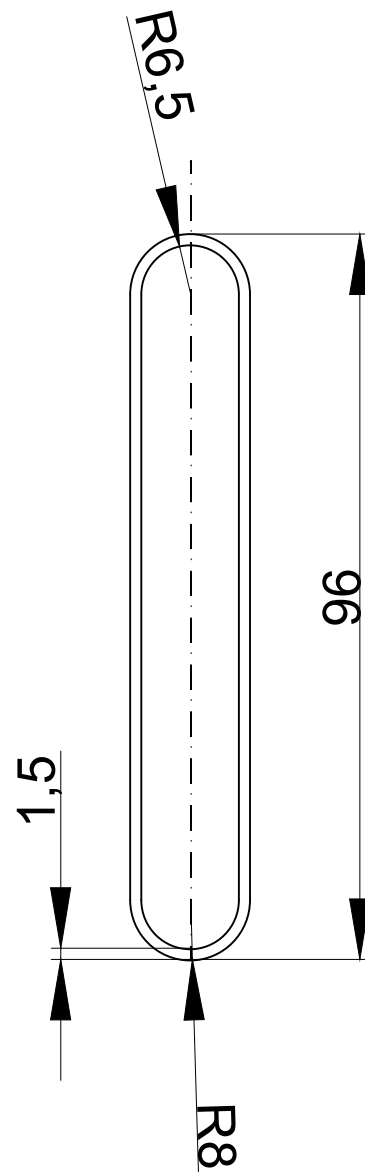
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| GRUPO PA                                | GRUPO PL         | ID DEL DIBUJO     | UNIDADES   |
| PA 07                                   | PL 19            | 1.5               | MM         |
| PLATAFORMA BASE: Estator y Eje Motor    |                  |                   | ESCALA 2:1 |
| INGENIERO/A RESPONSABLE: Daniel Fuentes |                  |                   |            |
| JEFE AL CARGO: Marcos Menendez          |                  |                   |            |
| MATERIAL PRINCIPAL                      | OTROS MATERIALES | NORMATIVA SEGUIDA |            |
| ALUMINIO                                | COBRE            | UNE 1-032-82      |            |



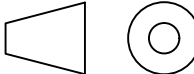


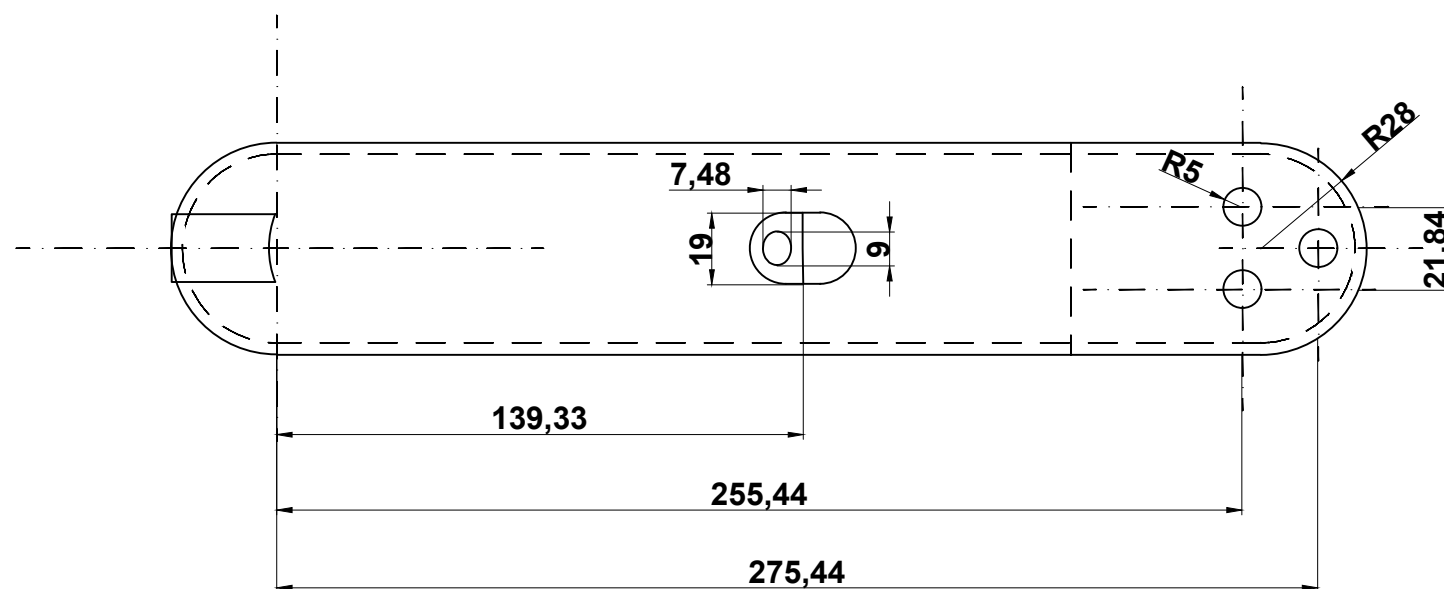
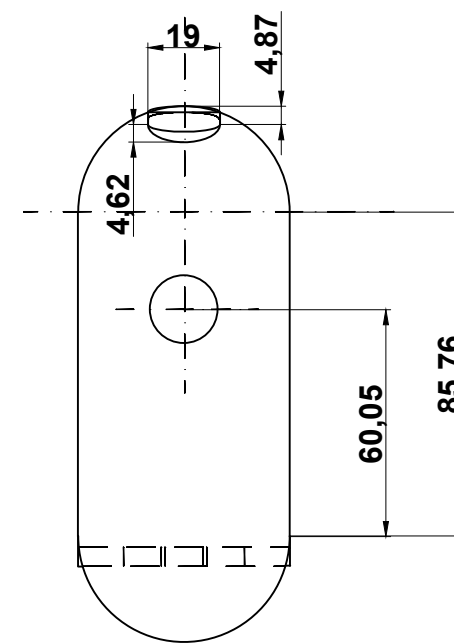
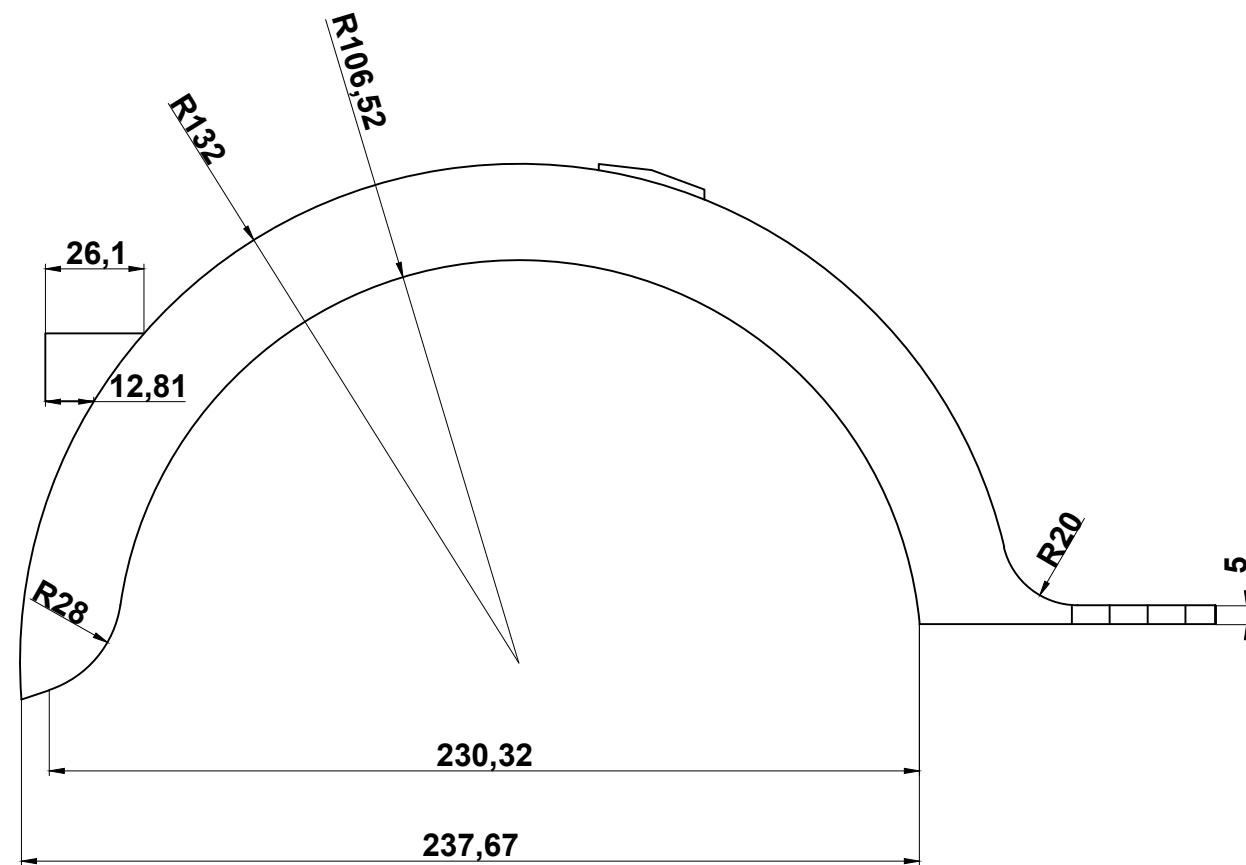
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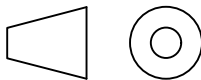
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| PA 07                                               | PL 19            | 1.6               | MM                                                                                    |
| HORQUILLA DELANTERA                                 |                  |                   | ESCALA 1:3                                                                            |
| INGENIERO RESPONSABLE:Miguel Ángel Rodríguez Ortega |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez Prieto               |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                  | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                            | ACERO            | UNE 1-032-82      |                                                                                       |

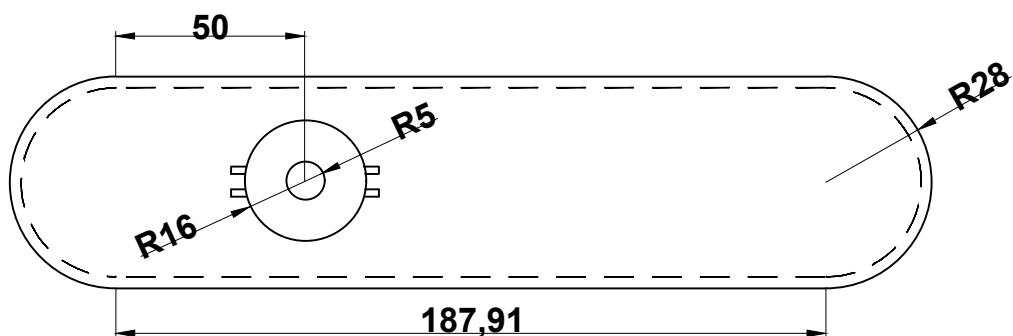
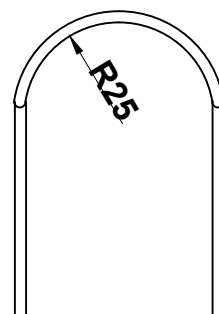
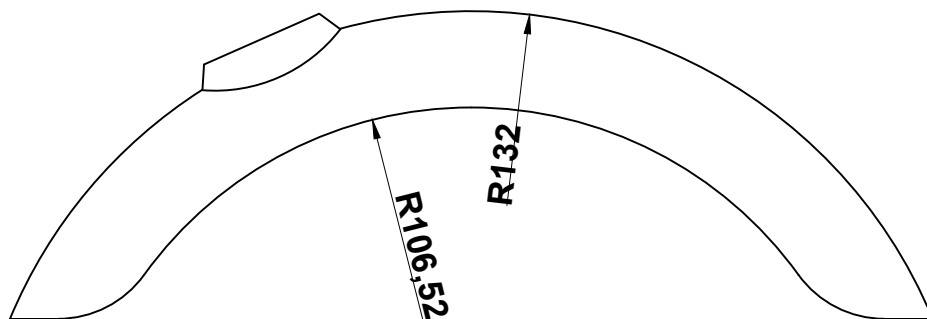


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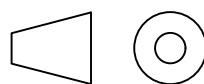
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| PA 07                                   | PL 19            | 1.7               | MM                                                                                    |
| EJE TRASERO Y ESPACIADORES              |                  |                   | ESCALA: 1:1                                                                           |
| INGENIERO/A RESPONSABLE: DANIEL SECADES |                  |                   |  |
| JEFE AL CARGO: MARCOS MENÉNDEZ          |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                      | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| PLÁSTICO                                | GOMA             | UNE 1-032-82      |                                                                                       |

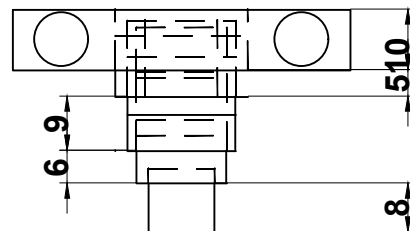
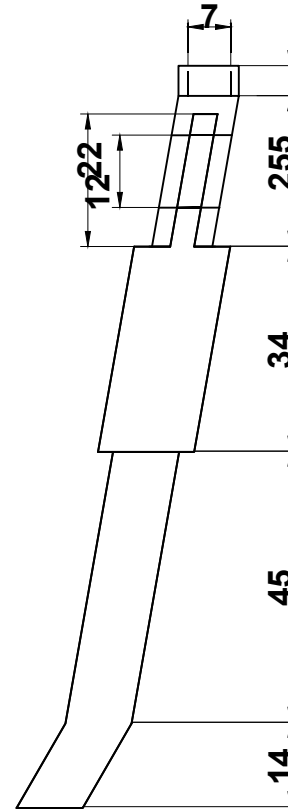
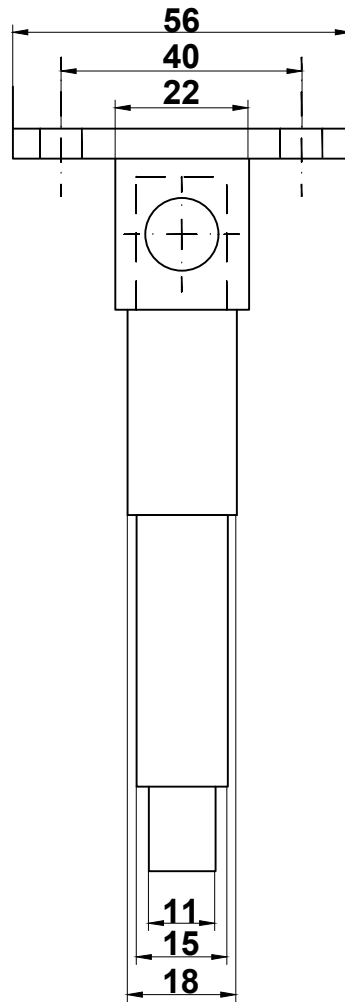


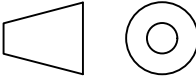
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| PA 07                                       | PL 19            | 1.8               | MM                                                                                    |
| GUARDABARROS TRASERO Y LUZ                  |                  |                   | ESCALA 1:2                                                                            |
| INGENIERO/A RESPONSABLE: Jacobo Silva Ridao |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet    |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                          | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| PLÁSTICO                                    | ACERO            | UNE 1-032-82      |                                                                                       |

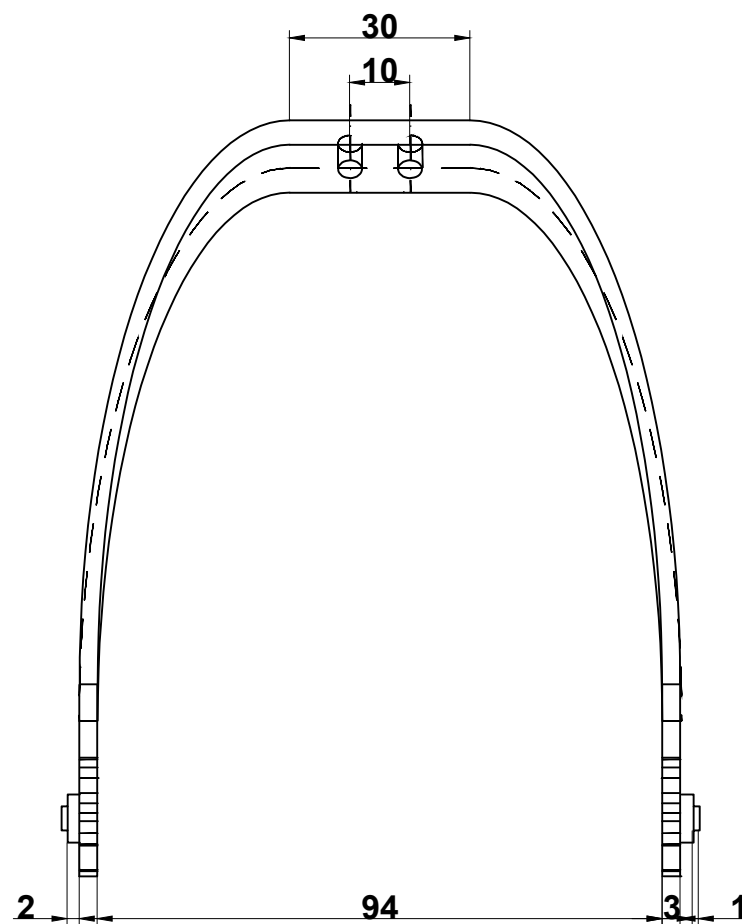
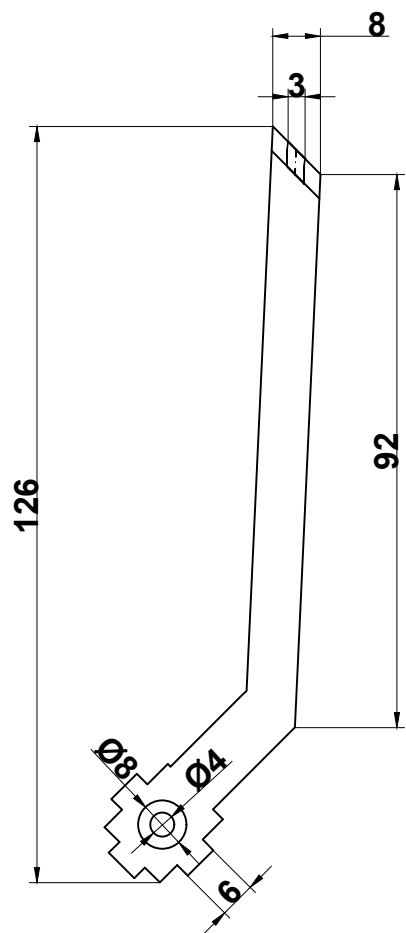


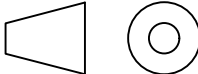
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| PA 07                                          | PL 19            | 1.9               | MM                                                                                    |
| GUARDABARROS DELANTERO                         |                  |                   | ESCALA 1:4                                                                            |
| INGENIERO/A RESPONSABLE: Claudiu Ricardo Barbu |                  |                   |  |
| JEFE AL CARGO: Marcos Menendez Prieto          |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                             | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| PLASTICO                                       | PLASTICO         | UNE 1-032-82      |                                                                                       |

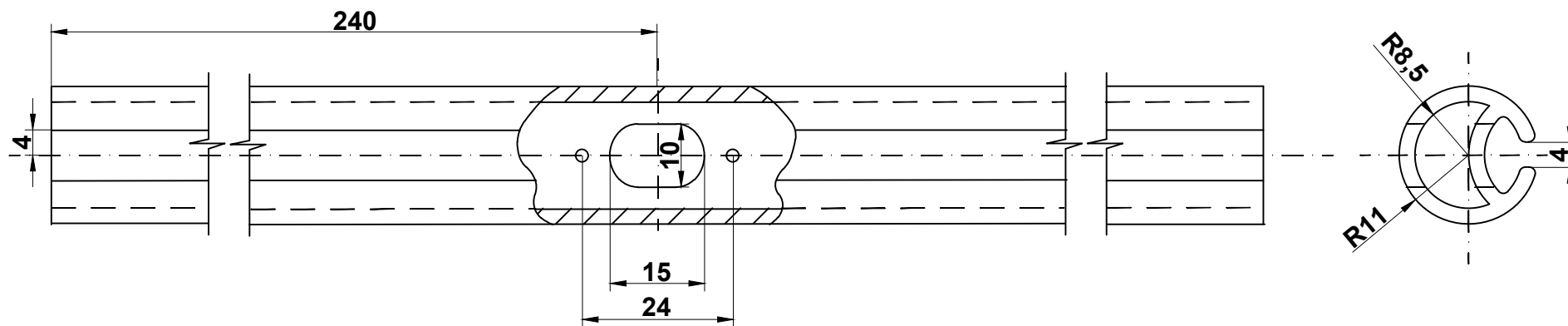


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| UNIVERSIDAD DE OVIEDO                           |                  |                   | PLANO 1 de 2                                                                          |
| GRUPO PA                                        | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                           | PL 19            | 1.10              | MM                                                                                    |
| CABALLETE Y TIRANTE                             |                  |                   | ESCALA 1:5                                                                            |
| INGENIERO/A RESPONSABLE: Marcos Menéndez Prieto |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez Prieto           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                              | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                        | ACERO            | UNE 1-032-82      |                                                                                       |



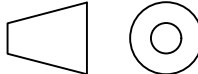
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| GRUPO PA                                        | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                           | PL 19            | 1.10              | MM                                                                                    |
| CABALLETE Y TIRANTE                             |                  |                   | ESCALA 1:5                                                                            |
| INGENIERO/A RESPONSABLE: Marcos Menéndez Prieto |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez Prieto           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                              | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                        | ACERO            | UNE 1-032-82      |                                                                                       |

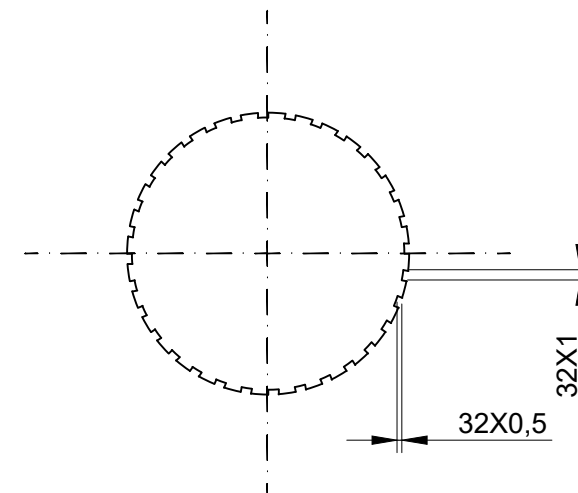
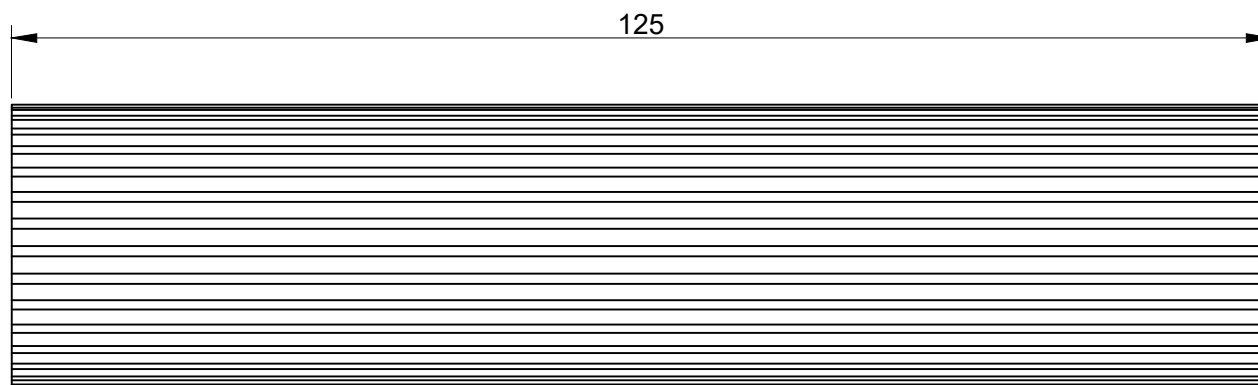


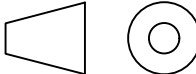


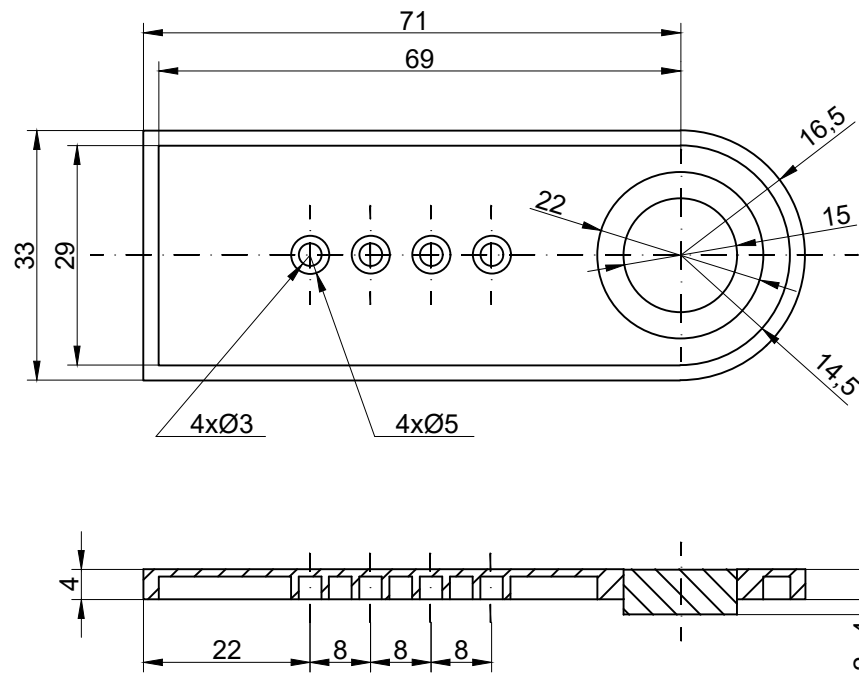
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PLANO 1 de 2

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| UNIVERSIDAD DE OVIEDO                 |                  |                   | PLANO 1 de 2                                                                          |
| GRUPO PA                              | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                 | PL 20            | 2.2               | MM                                                                                    |
| BARRA TRANSVERSAL Y EMPUÑADURAS       |                  |                   | ESCALA 1:0.75                                                                         |
| INGENIERO/A RESPONSABLE: ANDREA LÓPEZ |                  |                   |  |
| JEFE AL CARGO: ELOY SANTOS            |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                    | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| TPE                                   | CAUCHO SINTÉTICO | UNE 1-032-82      |                                                                                       |

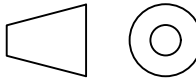


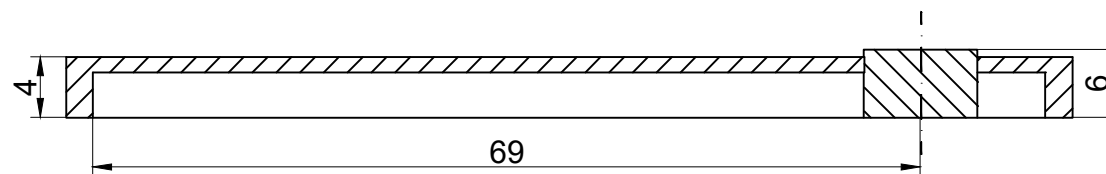
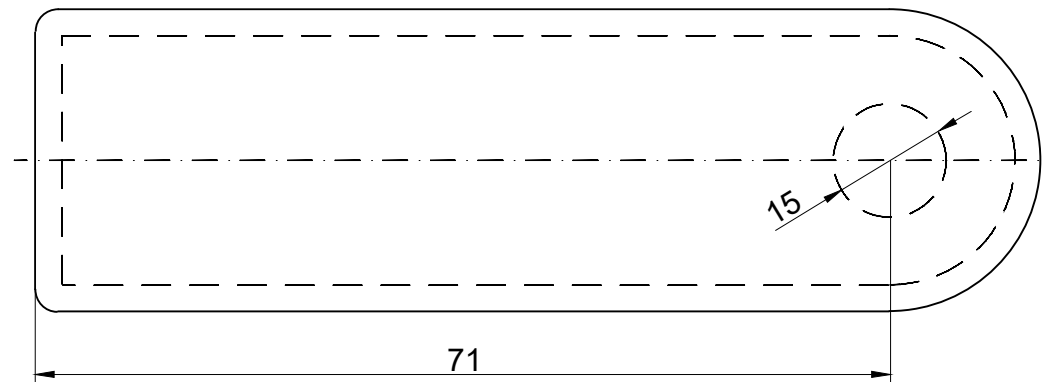
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| UNIVERSIDAD DE OVIEDO                 |                  |                   | PLANO 2 de 2                                                                          |
| GRUPO PA                              | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                 | PL 20            | 2.2               | MM                                                                                    |
| BARRA TRANSVERSAL Y EMPUÑADURAS       |                  |                   | ESCALA 1:0.75                                                                         |
| INGENIERO/A RESPONSABLE: ANDREA LÓPEZ |                  |                   |  |
| JEFE AL CARGO: ELOY SANTOS            |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                    | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| TPE                                   | CAUCHO SINTÉTICO | UNE 1-032-82      |                                                                                       |



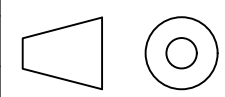
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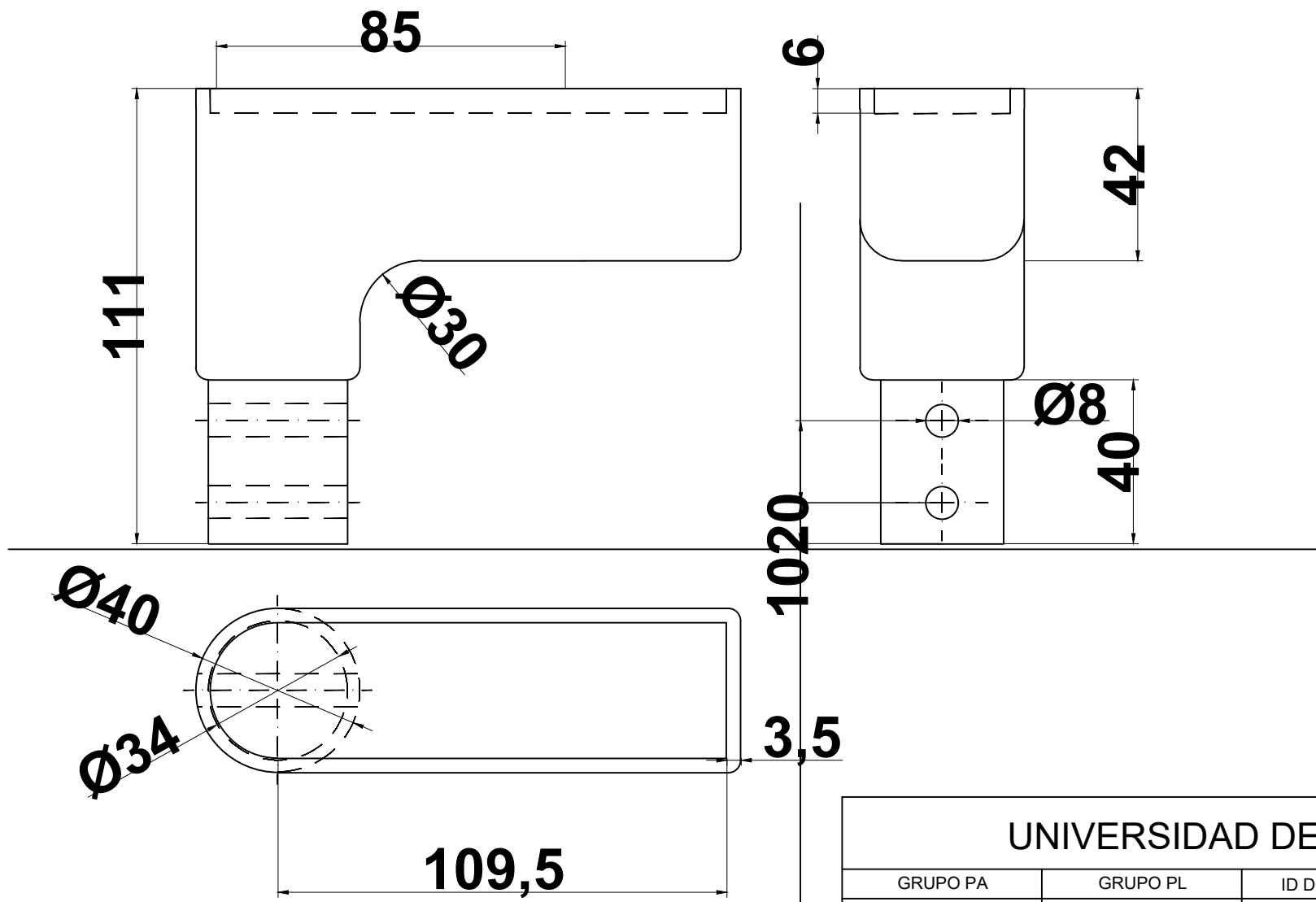
PLANO 1 de 2

|                                          |                  |                   |                                                                                       |
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| UNIVERSIDAD DE OVIEDO                    |                  |                   | PLANO 1 de 2                                                                          |
| GRUPO PA                                 | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                    | PL 20            | 2.3               | MM                                                                                    |
| CUBIERTA SUPERIOR DISPLAY                |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Lorenzo Sanchez |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                       | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                 | ACERO            | UNE 1-032-82      |                                                                                       |

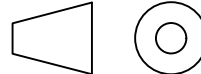


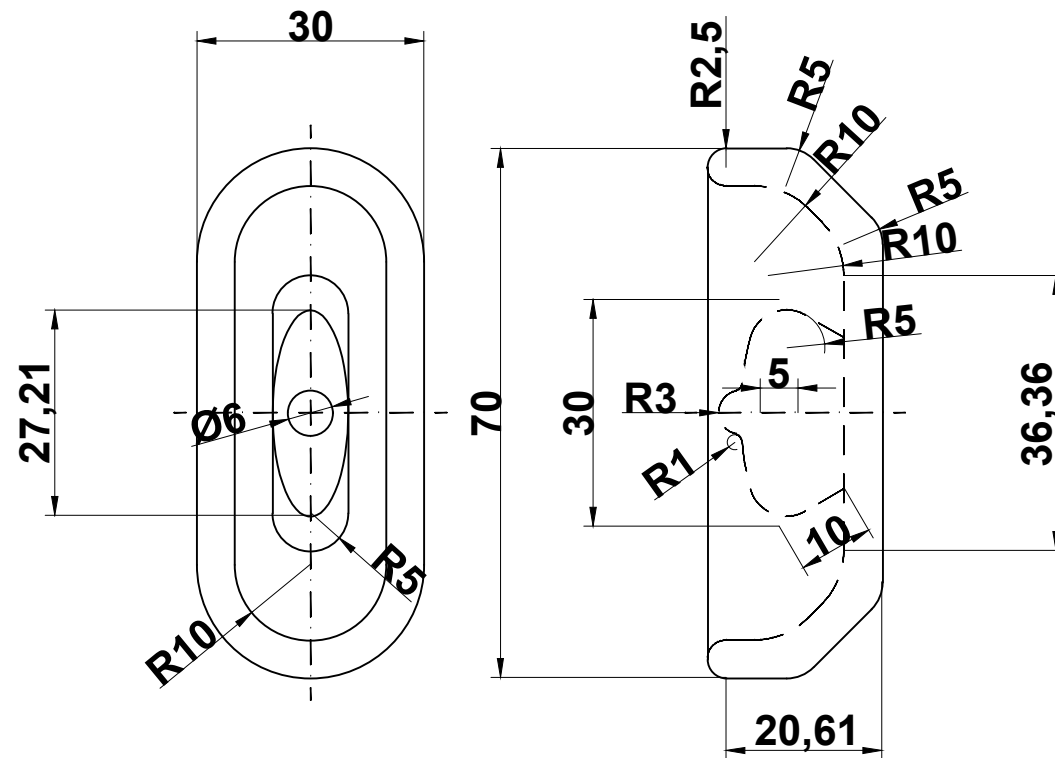
## UNIVERSIDAD DE OVIEDO

|                                          |                  |                   |                                                                                       |
|------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                 | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                    | PL 20            | 2.3               | MM                                                                                    |
| CUBIERTA SUPERIOR DISPLAY                |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Lorenzo Sanchez |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                       | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                 | ACERO            | UNE 1-032-82      |                                                                                       |

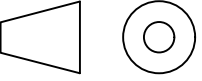


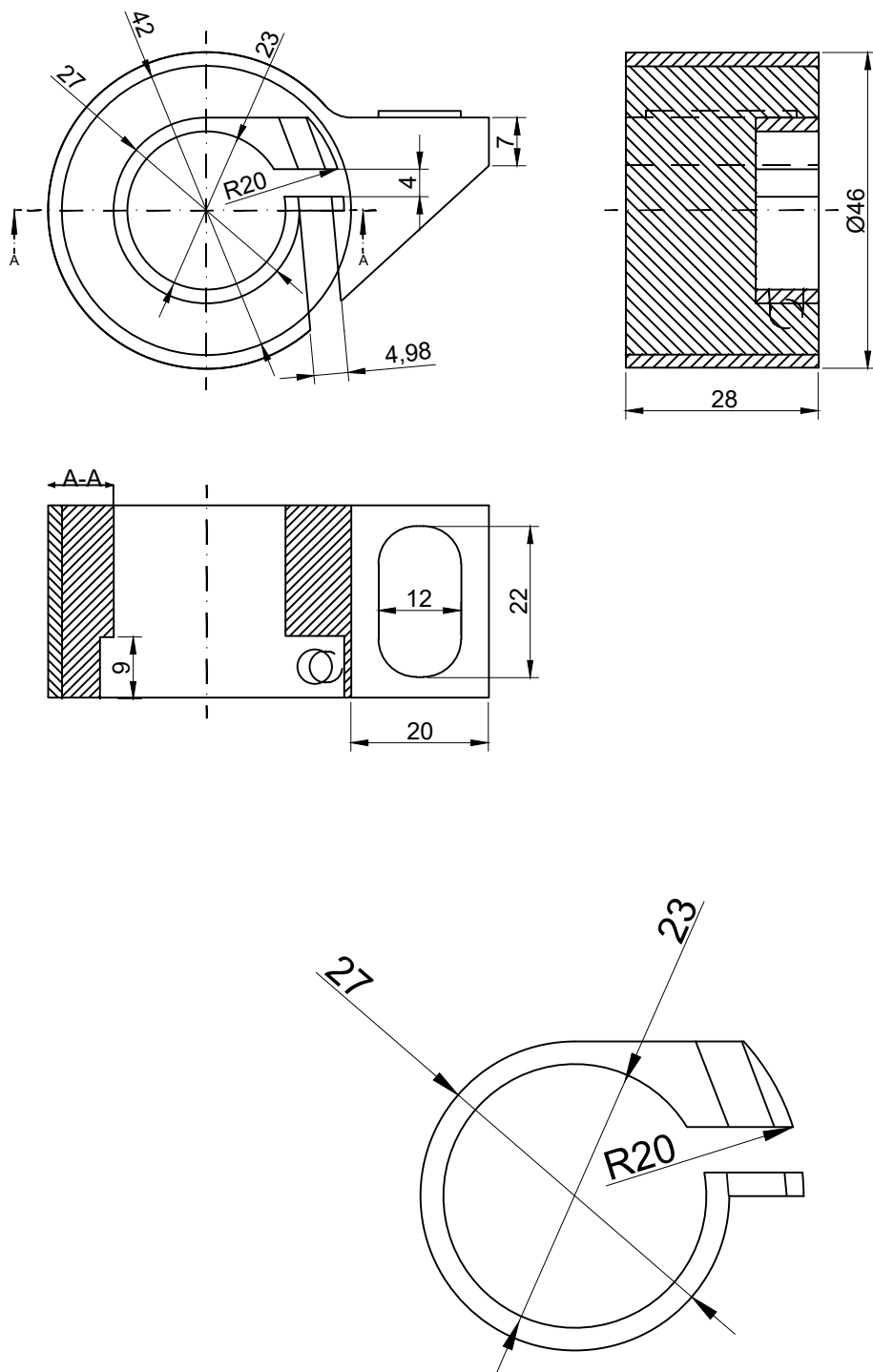
## UNIVERSIDAD DE OVIEDO

|                                                  |                  |                   |                                                                                       |
|--------------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                         | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                            | PL 20            | 2.4               | MM                                                                                    |
| BASE INFERIOR DISPLAY                            |                  |                   | ESCALA 2:3                                                                            |
| INGENIERO/A RESPONSABLE: Sebastián Aguilar López |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet         |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                               | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                         | CAUCHO           | UNE 1-032-82      |                                                                                       |



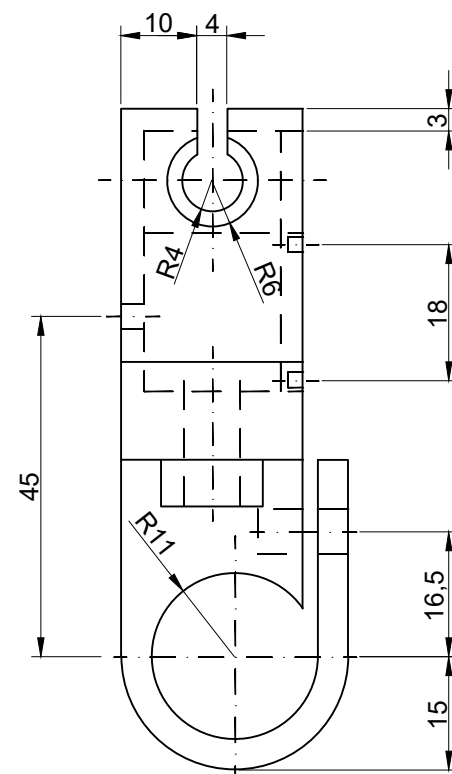
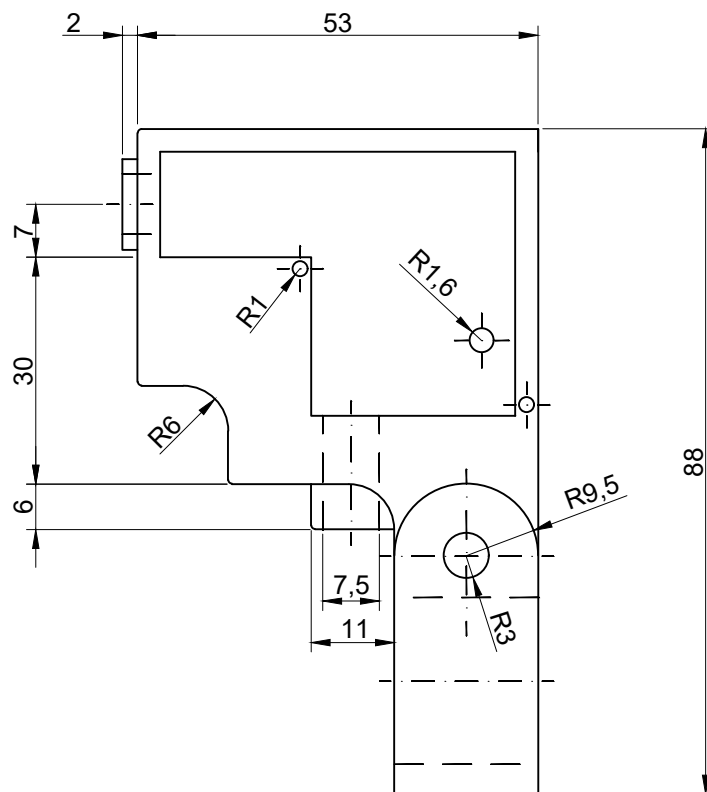
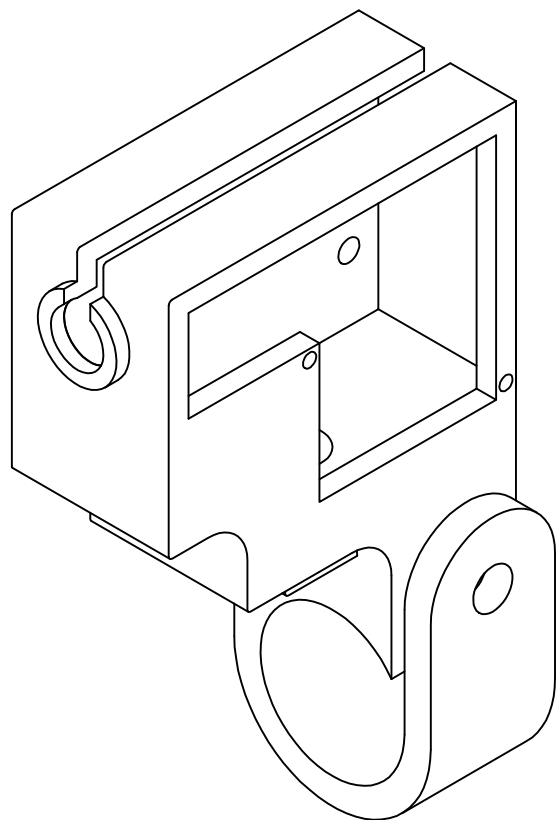
## UNIVERSIDAD DE OVIEDO

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| GRUPO PA                                           | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                              | PL 20            | 2.5               | MM                                                                                    |
| CARCASA FARO Y LED                                 |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Eloy Santos de Tourdonnet |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                           | POLICARBONATO    | UNE 1-032-82      |                                                                                       |



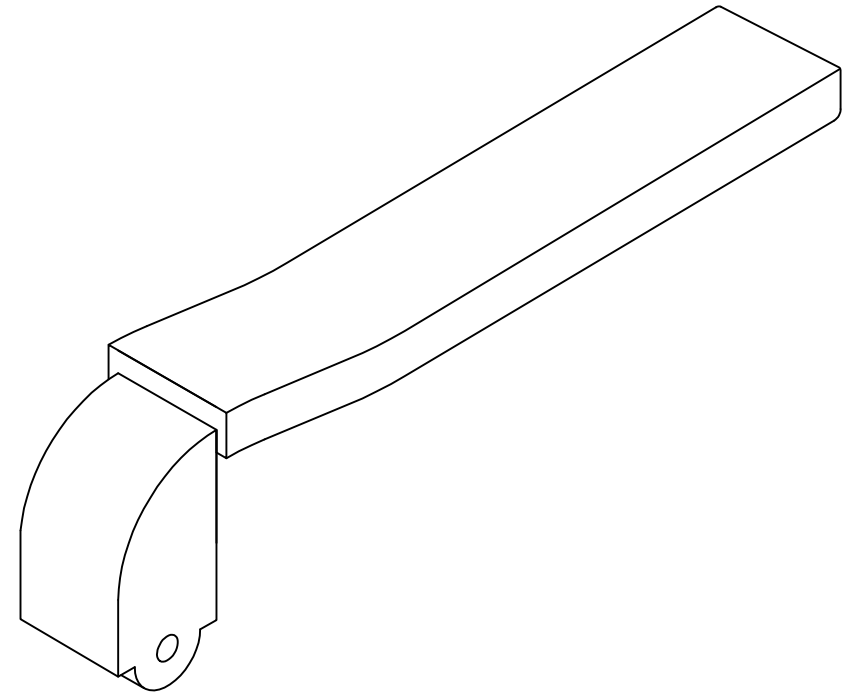
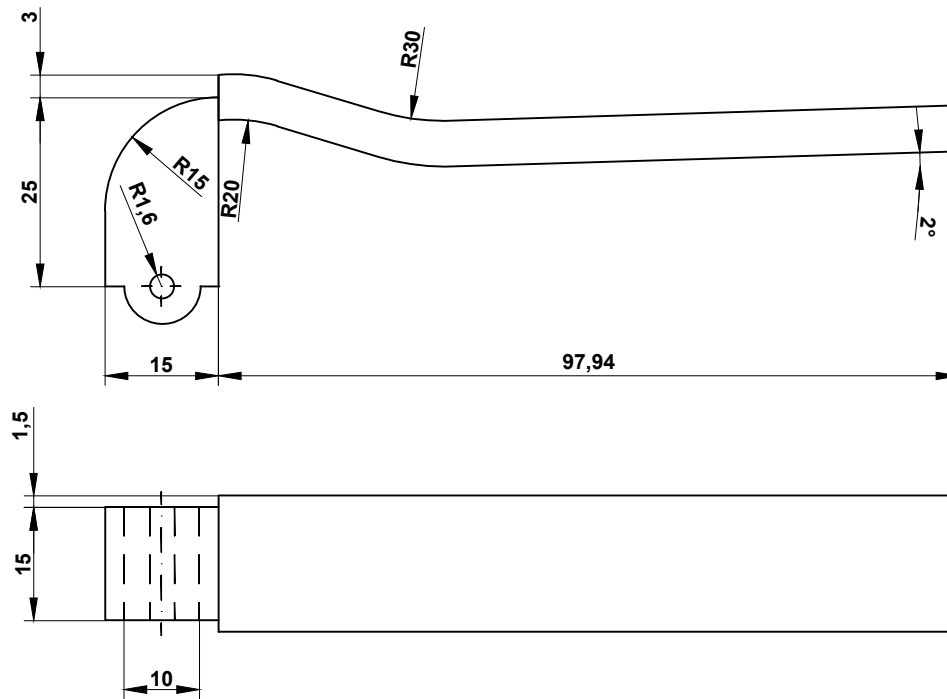
## UNIVERSIDAD DE OVIEDO

| GRUPO PA                                  | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
|-------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| PA 07                                     | PL 20            | 2.6               | MM                                                                                    |
| ACELERADOR                                |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Iyán Berjano Ces |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet  |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                        | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                  | PLÁSTICO         | UNE 1-032-82      |                                                                                       |

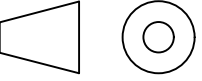


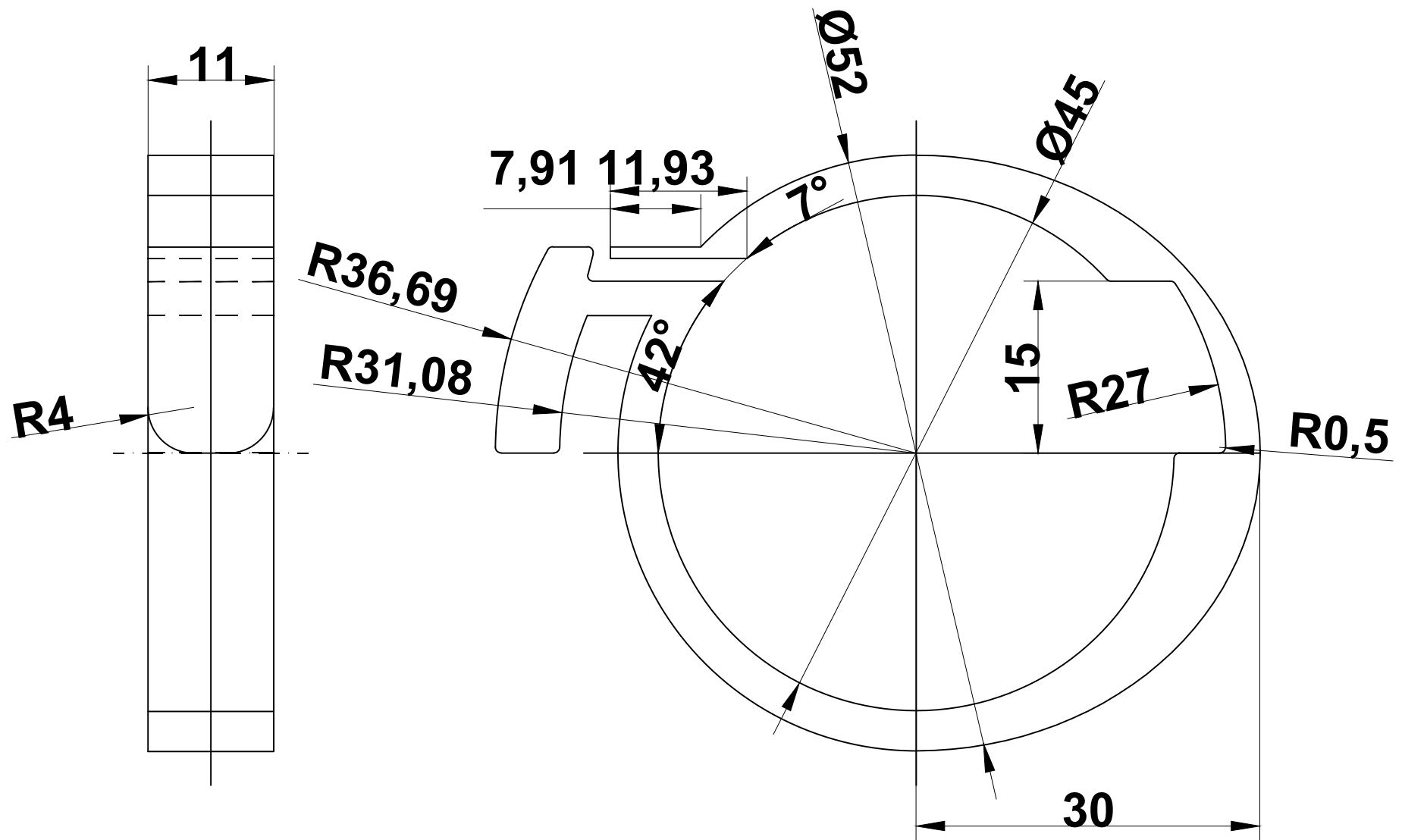
## UNIVERSIDAD DE OVIEDO

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| GRUPO PA                                      | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                                                                                                                    |
| PA 07                                         | PL 21            | 2.8               | MM                                                                                                                                                                          |
| CUERPO MANETA                                 |                  |                   | ESCALA: 1:1                                                                                                                                                                 |
| INGENIERO/A RESPONSABLE: Ian Morerías Matínez |                  |                   |   |
| JEFE AL CARGO: Hugo Ojeda Viedma              |                  |                   |                                                                                                                                                                             |
| MATERIAL PRINCIPAL                            | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                                                                                                             |
| ACERO                                         | ACERO            | UNE 1-032-82      |                                                                                                                                                                             |

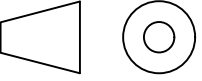


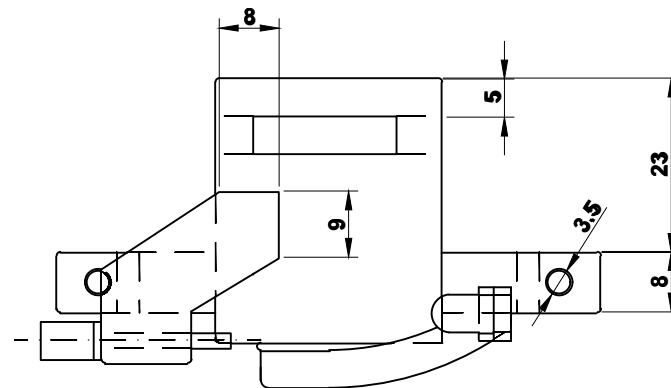
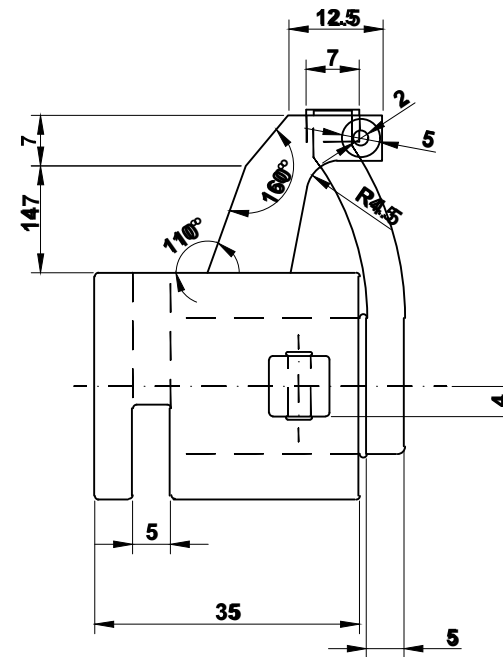
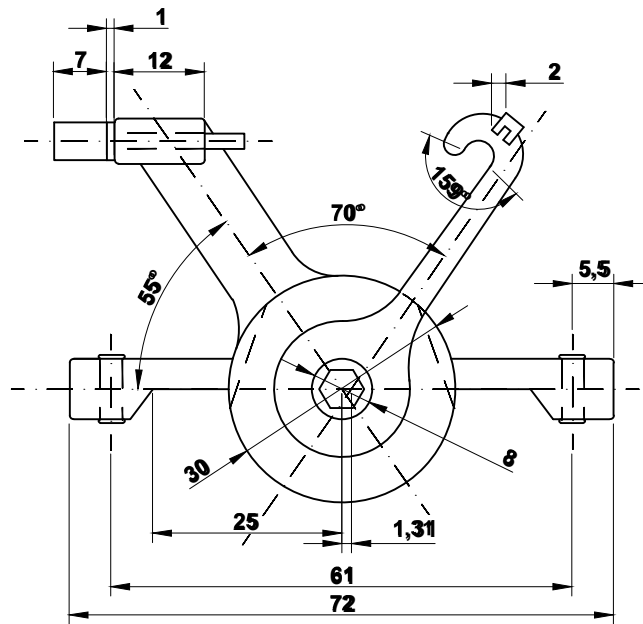
## UNIVERSIDAD DE OVIEDO

|                                               |                  |                   |                                                                                       |
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| GRUPO PA                                      | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                         | PL 21            | 2.8               | MM                                                                                    |
| PALANCA DE FRENO                              |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Ian Morerías Matínez |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma              |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                            | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ACERO                                         | ACERO            | UNE 1-032-82      |                                                                                       |

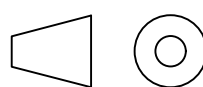


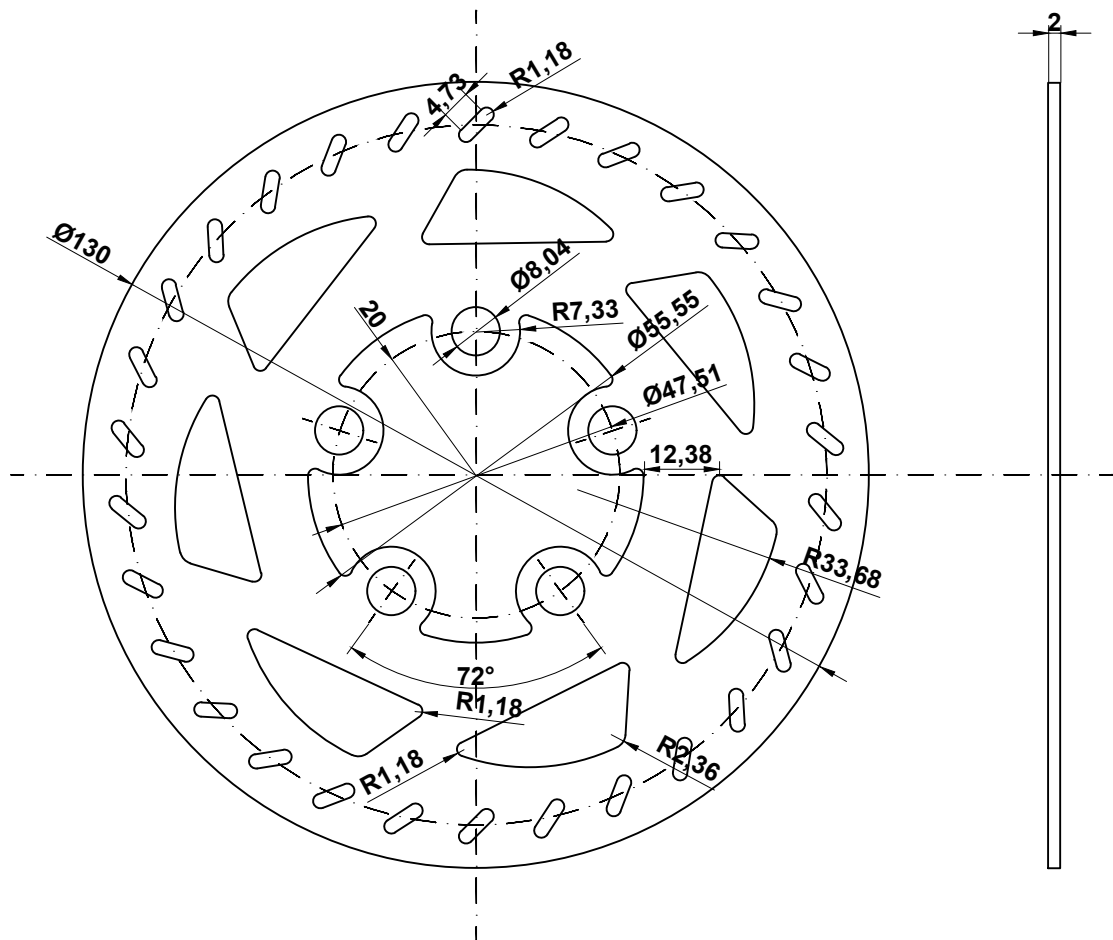
## UNIVERSIDADDE OVIEDO

|                                          |                  |                   |                                                                                       |
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| GRUPO PA                                 | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                    | PL 20            | 2.9               | MM                                                                                    |
| ANILLO DE SEGURIDAD                      |                  |                   | ESCALA 2:1                                                                            |
| INGENIERO/A RESPONSABLE: ANDREA LÓPEZ    |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                       | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                 | -                | UNE 1-032-82      |                                                                                       |

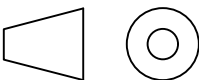


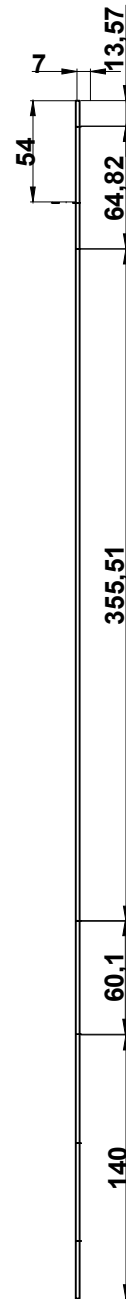
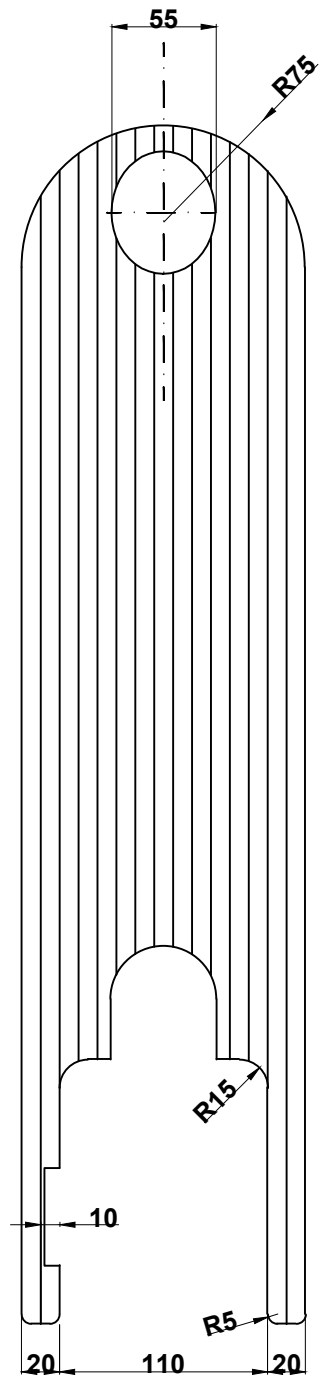
## UNIVERSIDAD DE OVIEDO

|                                         |                  |                   |                                                                                       |
|-----------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                   | PL 20            | 2.10              | MM                                                                                    |
| PINZA DE FRENO                          |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Rubén Iordache |                  |                   |  |
| JEFE AL CARGO: Nombre de ejemplo        |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                      | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                | ACERO            | UNE 1-032-82      |                                                                                       |



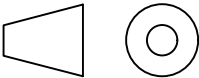
## UNIVERSIDAD DE OVIEDO

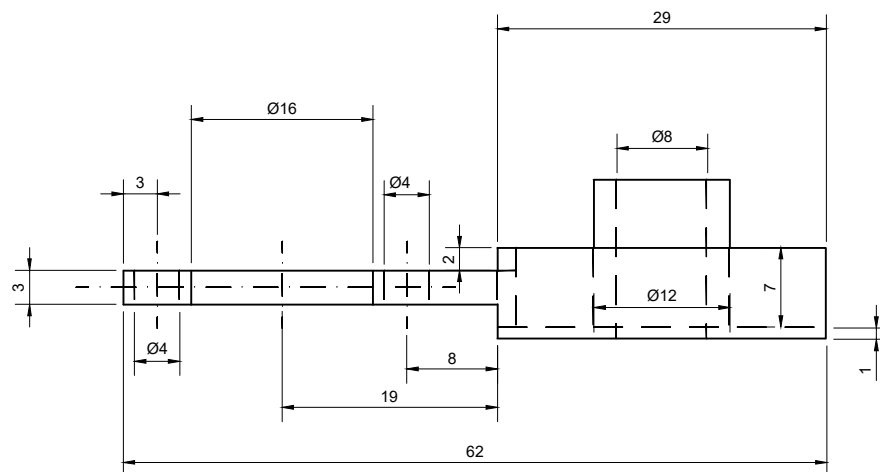
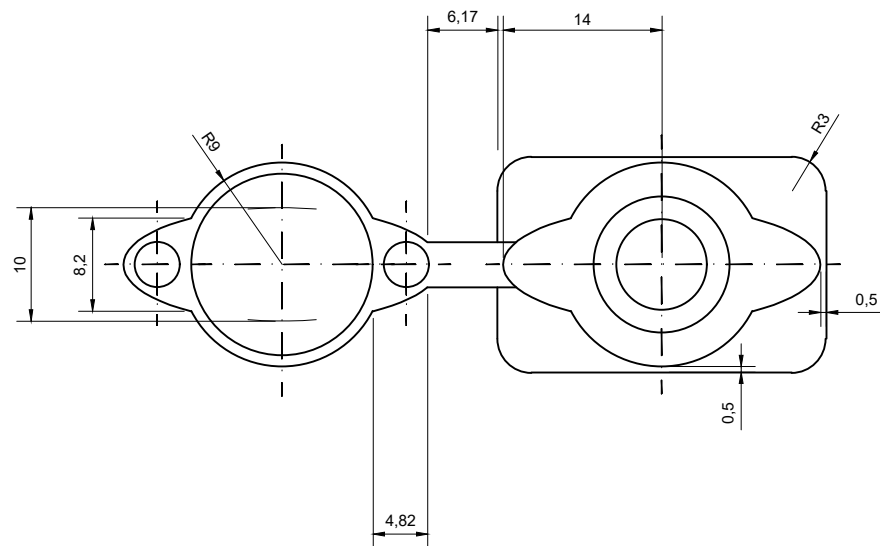
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| PA 07                                              | PL 20            | 2.11              | MM                                                                                    |
| DISCO DE FRENO                                     |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Eloy Santos de Tourdonnet |                  |                   |  |
| JEFE AL CARGO: Eloy Santos de Tourdonnet           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ACERO                                              | -                | UNE 1-032-82      |                                                                                       |



# UNIVERSIDAD DE OVIEDO

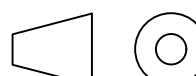
PLANO 1 de 2

|                                            |                  |                   |                                                                                       |
|--------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| UNIVERSIDAD DE OVIEDO                      |                  |                   | PLANO 1 de 2                                                                          |
| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                      | PL 21            | 3.1               | MM                                                                                    |
| GOMA ANTIDESLIZANTE Y TAPA DE CARGA        |                  |                   | ESCALA 1:4                                                                            |
| INGENIERO/A RESPONSABLE: Iván Dou González |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| SILICONA                                   | TPE              | UNE 1-032-82      |                                                                                       |

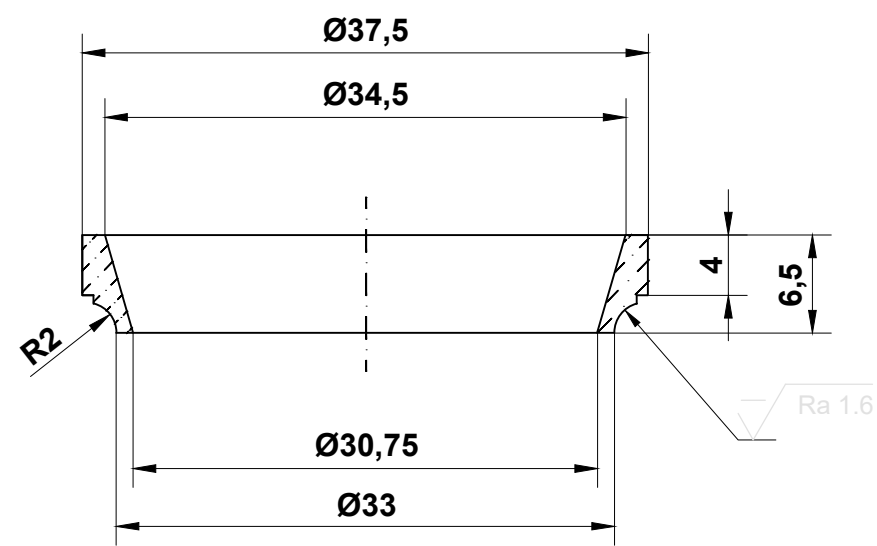


# UNIVERSIDAD DE OVIEDO

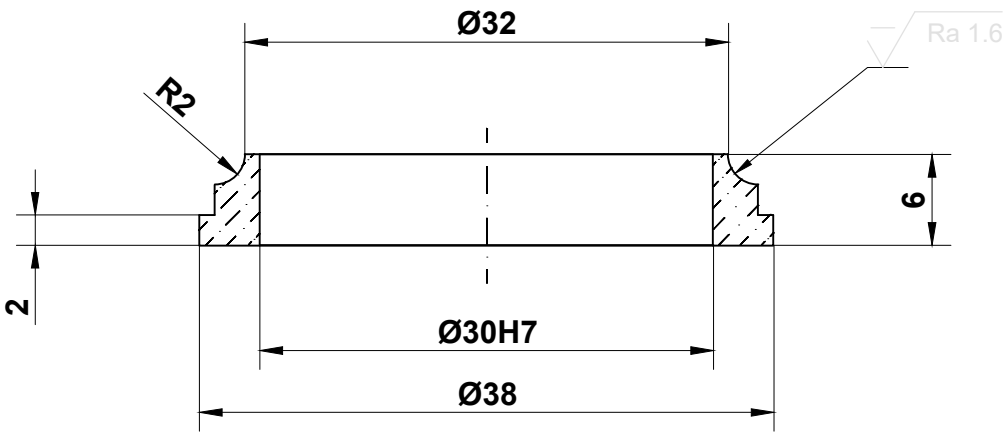
PLANO 2 de 2

|                                            |                  |                   |                                                                                       |
|--------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                      | PL 21            | 3.1               | MM                                                                                    |
| GOMA ANTIDESLIZANTE Y TAPA DE CARGA        |                  |                   | ESCALA 1:4                                                                            |
| INGENIERO/A RESPONSABLE: Iván Dou González |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| SILICONA                                   | TPE              | UNE 1-032-82      |                                                                                       |

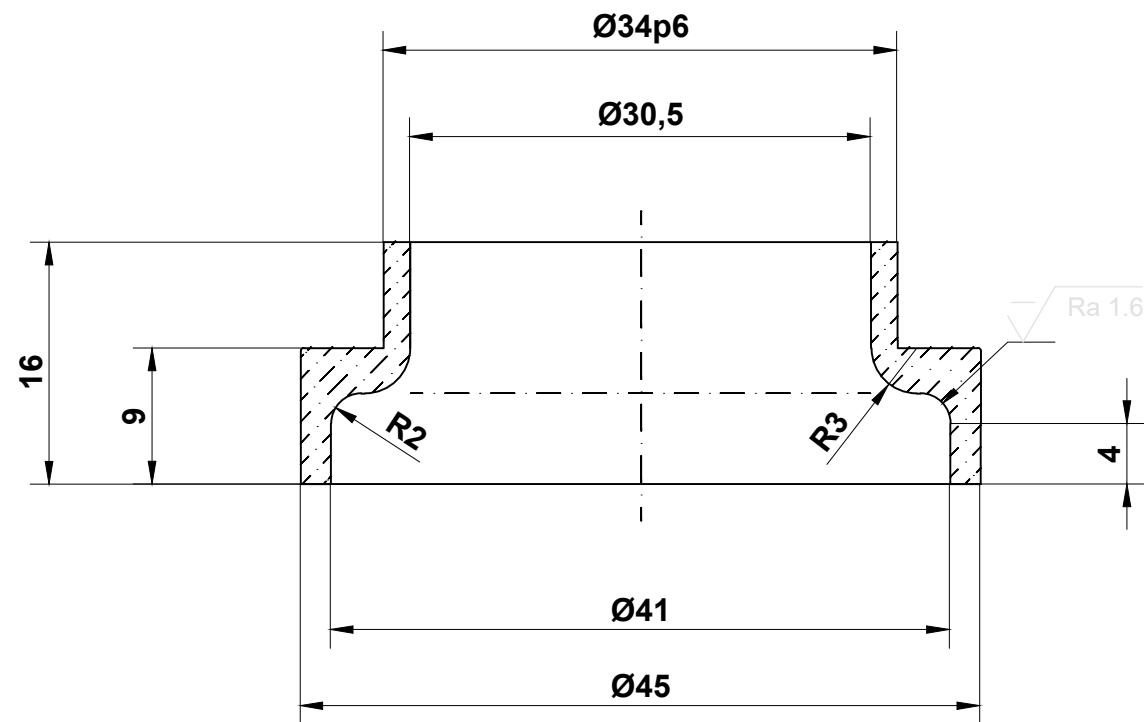
MARCA 5

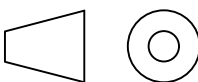


MARCA 8

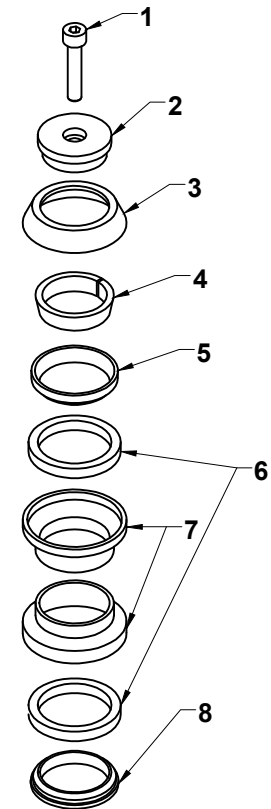
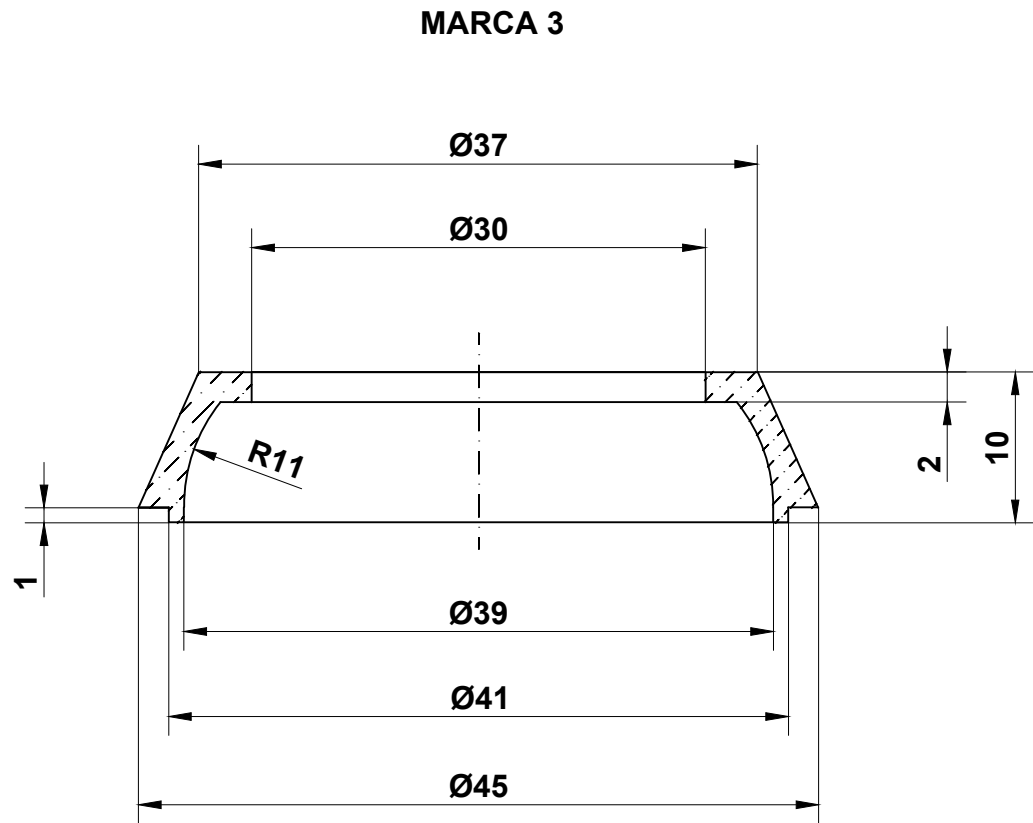
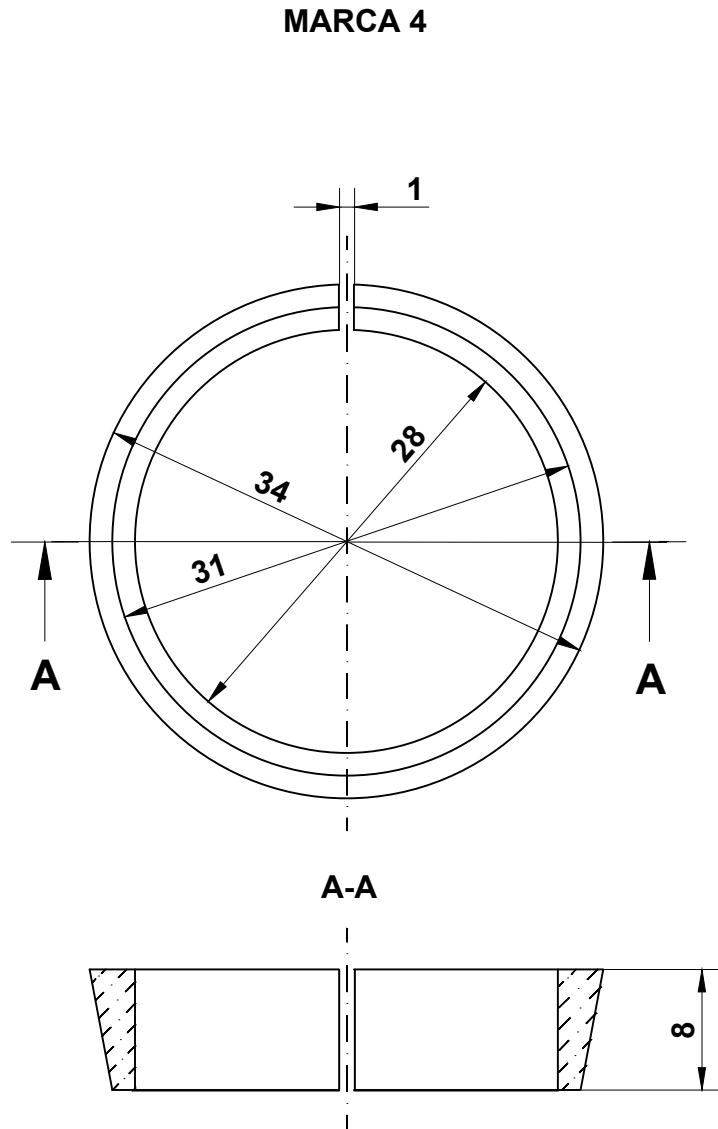
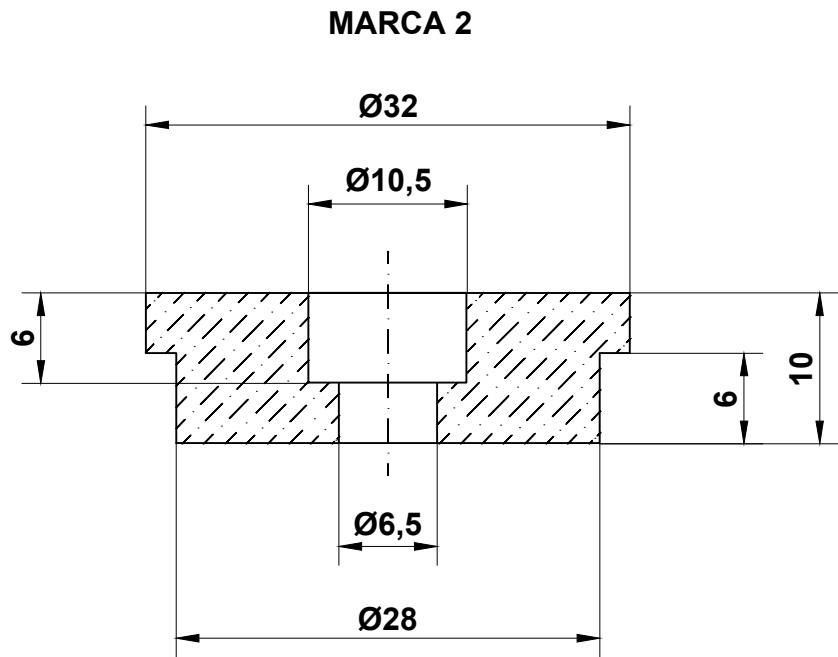


MARCA 7



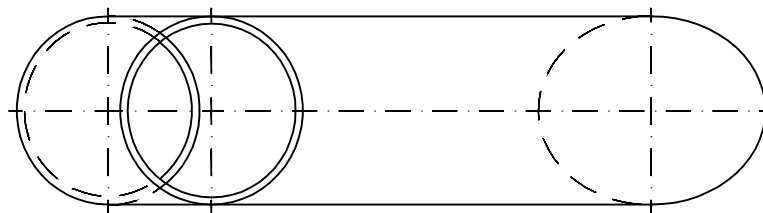
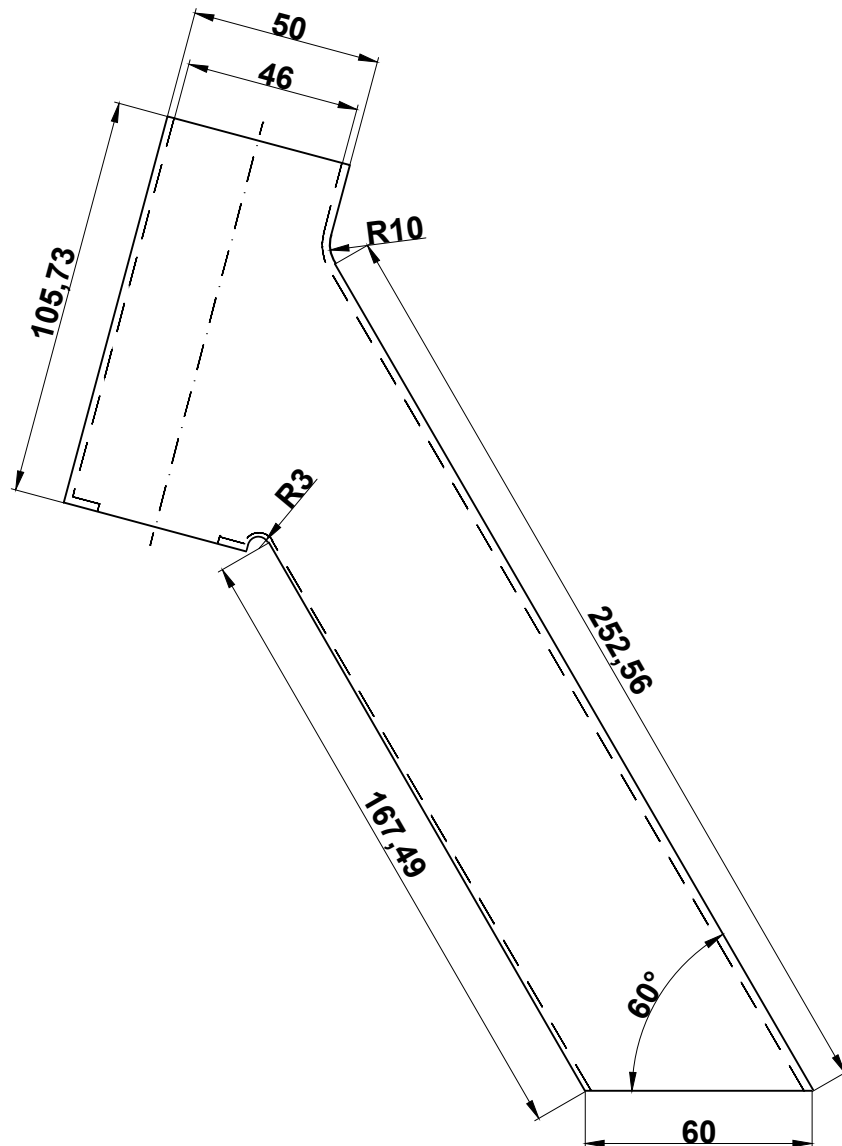
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|----------------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| UNIVERSIDAD DE OVIEDO                              |                  |                   | PLANO 1 de 2                                                                          |
| GRUPO PA                                           | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                              | PL 21            | 3.2               | MM                                                                                    |
| JUEGO DE DIRECCIÓN                                 |                  |                   | ESCALA 2:1                                                                            |
| INGENIERO/A RESPONSABLE: Francisco Villanueva Fdez |                  |                   |  |
| JEFE AL CARGO: Nombre de ejemplo                   |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                           | ACERO            | UNE 1-032-82      |                                                                                       |

TOLERANCIAS GENERALES: ISO 2768-m

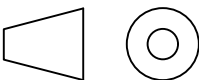


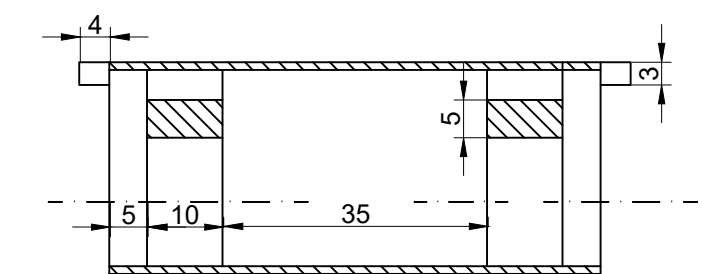
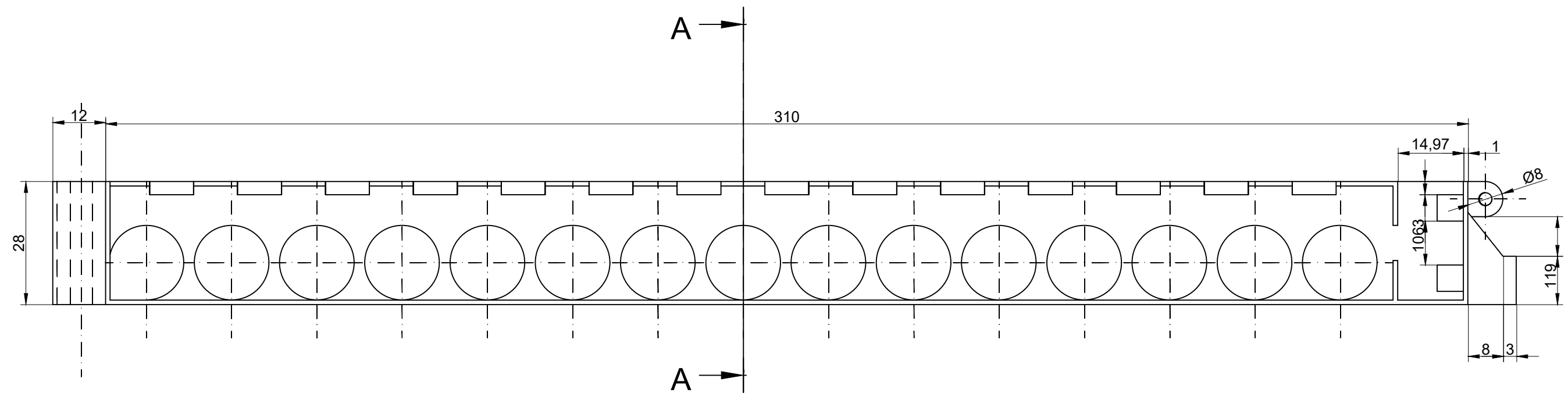
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|----------------------------------------------------|------------------|------------------------------------------------------------------|---------------|---------------|
| 8                                                  | 1                | Pista Inferior                                                   | EN ISO 683-17 | 100Cr6        |
| 7                                                  | 2                | Cazoletas                                                        | EN ISO 683-17 | 100Cr6        |
| 6                                                  | 2                | Jaula de bolas de direcci3n 1 1/8 pulgadas, 5/32 x 20, 40 x 32mm |               |               |
| 5                                                  | 1                | Pista Superior                                                   | EN ISO 683-17 | 100Cr6        |
| 4                                                  | 1                | Casquillo de ajuste                                              | EN ISO 683-17 | 100Cr6        |
| 3                                                  | 1                | Guardapolvos                                                     |               | 100Cr6 + NBR  |
| 2                                                  | 1                | Tapa Superior de Precarga                                        | EN ISO 683-17 | 100Cr6        |
| 1                                                  | 1                | TORNILLO ALLEN 8.8 ZINCADO M6 x30                                | ISO 4762      | Acero Zincado |
| POS.                                               | CANT.            | DESCRIPC3N                                                       | NORMA         | MATERIAL      |
| UNIVERSIDAD DE OVIEDO                              |                  |                                                                  |               | PLANO 2 de 2  |
| GRUPO PA                                           |                  | GRUPO PL                                                         | ID DEL DIBUJO | UNIDADES      |
| PA 07                                              |                  | PL 21                                                            | 3.2           | MM            |
| PLATAFORMA BASE                                    |                  |                                                                  |               | ESCALA 2:1    |
| INGENIERO/A RESPONSABLE: Francisco Villanueva Fdez |                  |                                                                  |               |               |
| JEFE AL CARGO: Nombre de ejemplo                   |                  |                                                                  |               |               |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA                                                |               |               |
| ACERO                                              | ACERO            | UNE 1-032-82                                                     |               |               |

TOLERANCIAS GENERALES: ISO 2768-m

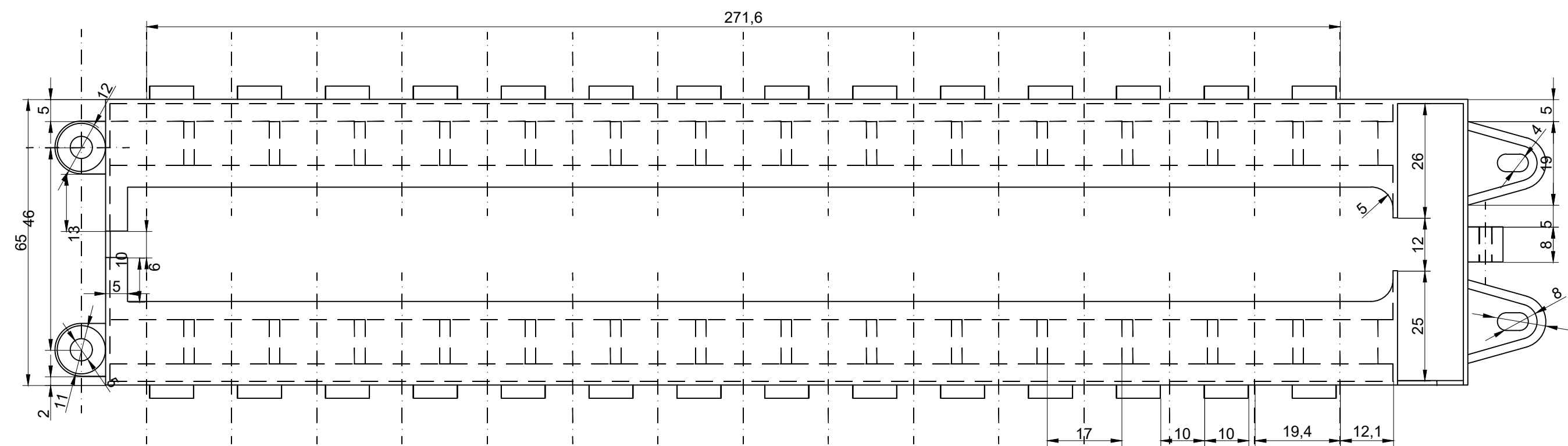


## UNIVERSIDAD DE OVIEDO

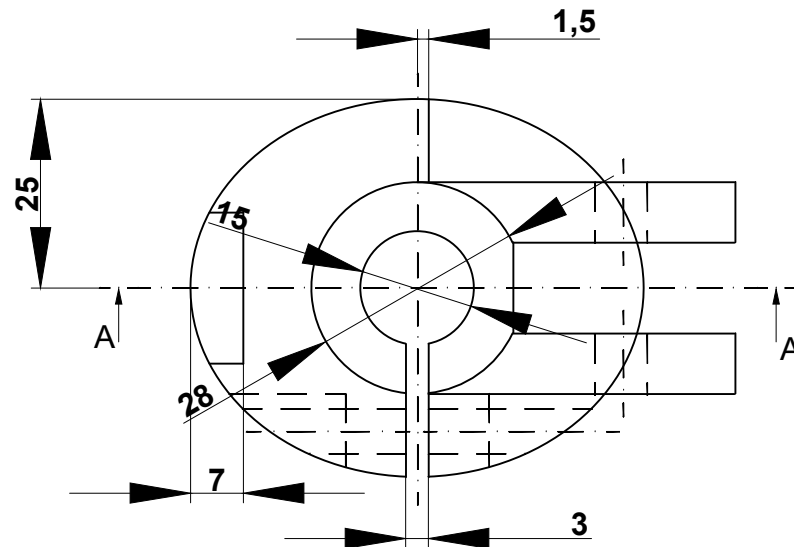
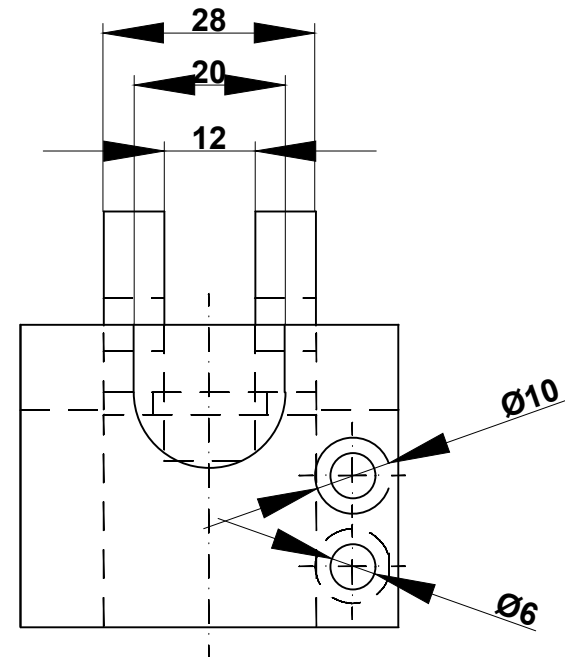
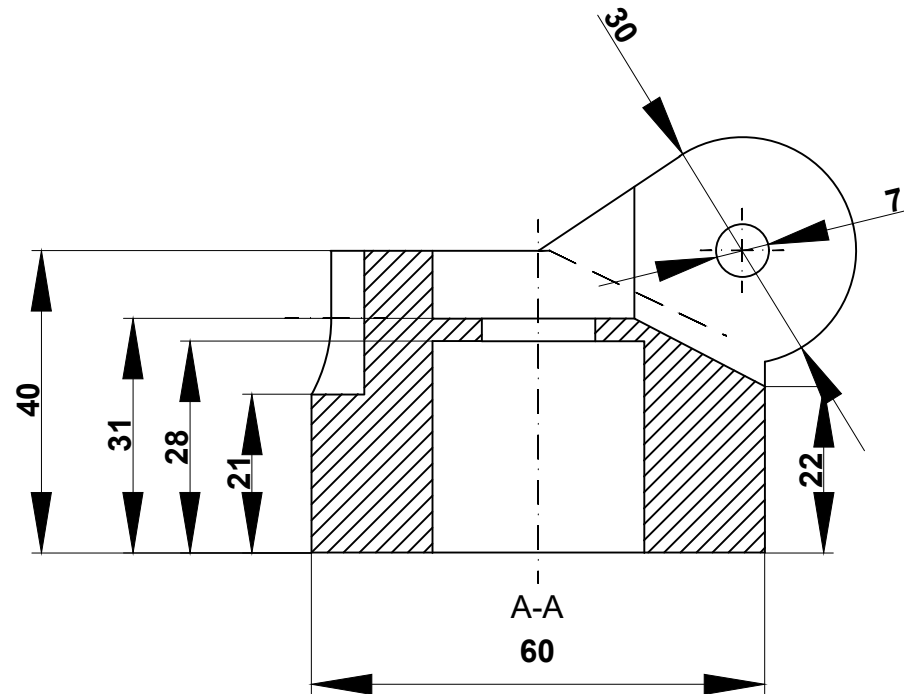
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| PA 07                                          | PL 21            | 3.3               | MM                                                                                    |
| CUELLO DE UNION                                |                  |                   | ESCALA 1:2                                                                            |
| INGENIERO/A RESPONSABLE: Ian Morerías Martínez |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma               |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                             | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ACERO                                          | ACERO            | UNE 1-032-82      |                                                                                       |



A-A

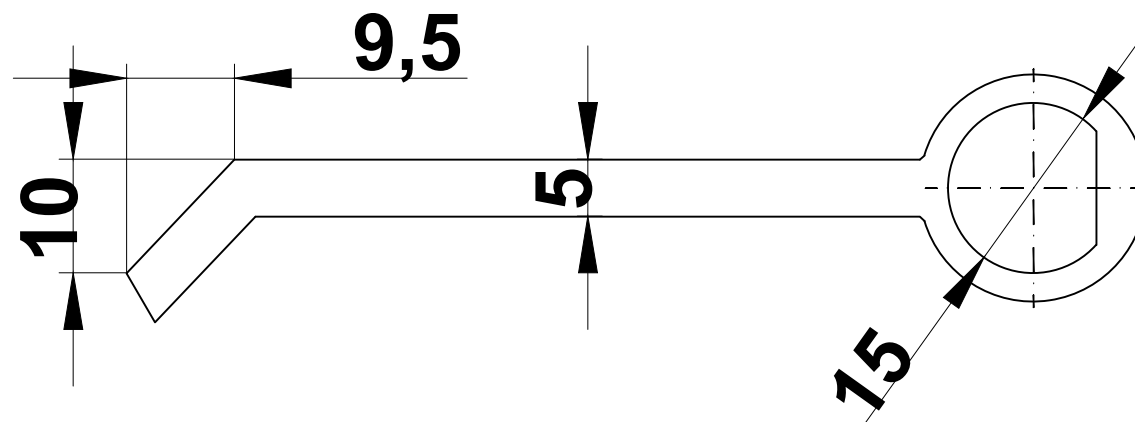
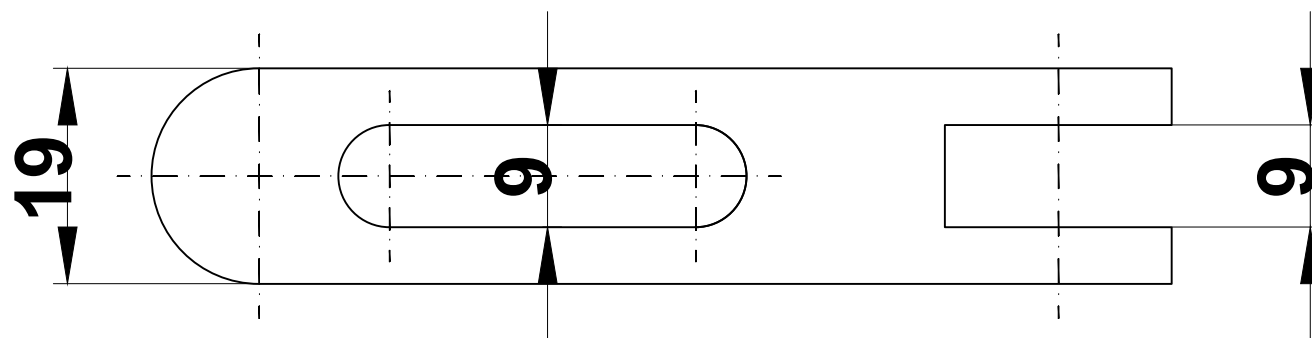


| UNIVERSIDAD DE OVIEDO                       |                  |                   |            |
|---------------------------------------------|------------------|-------------------|------------|
| GRUPO PA                                    | GRUPO PL         | ID DEL DIBUJO     | UNIDADES   |
| PA 07                                       | PL 21            | 3.4               | MM         |
| BATERÍA                                     |                  |                   | ESCALA 1:1 |
| INGENIERO/A RESPONSABLE: Vega Fernández Uña |                  |                   |            |
| JEFE AL CARGO: Hugo                         |                  |                   |            |
| MATERIAL PRINCIPAL                          | OTROS MATERIALES | NORMATIVA SEGUIDA |            |
| LITIO                                       | ALUMINIO         | UNE 1-032-82      |            |

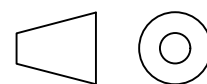


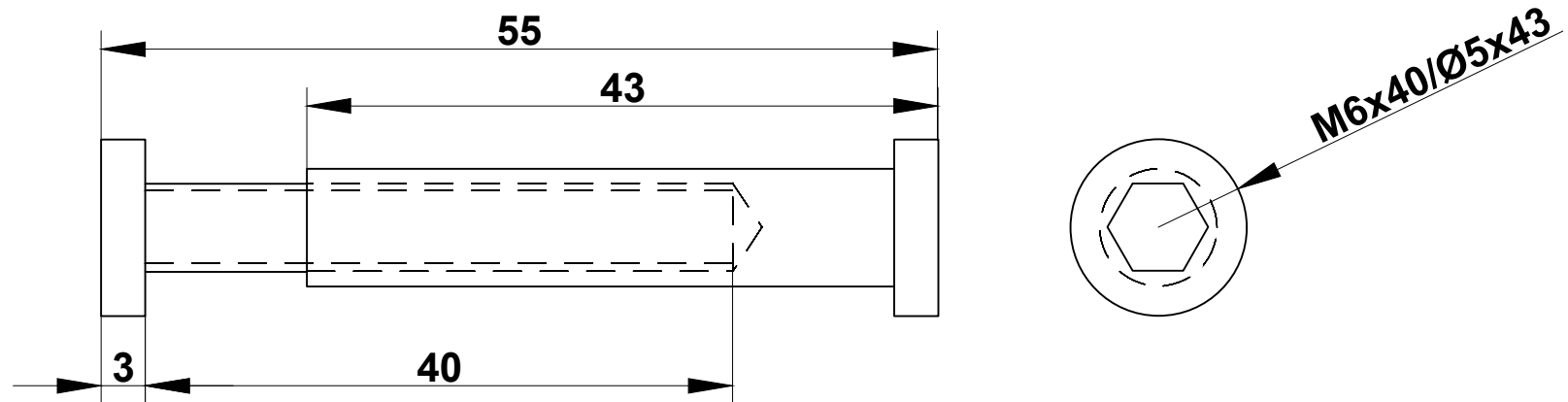
## UNIVERSIDAD DE OVIEDO

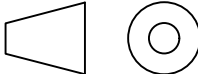
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|--------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                      | PL 21            | 3.5               | MM                                                                                    |
| BASE SISTEMA DE PLEGADO                    |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Hugo Ojeda Viedma |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                   | ACERO            | UNE 1-032-82      |                                                                                       |

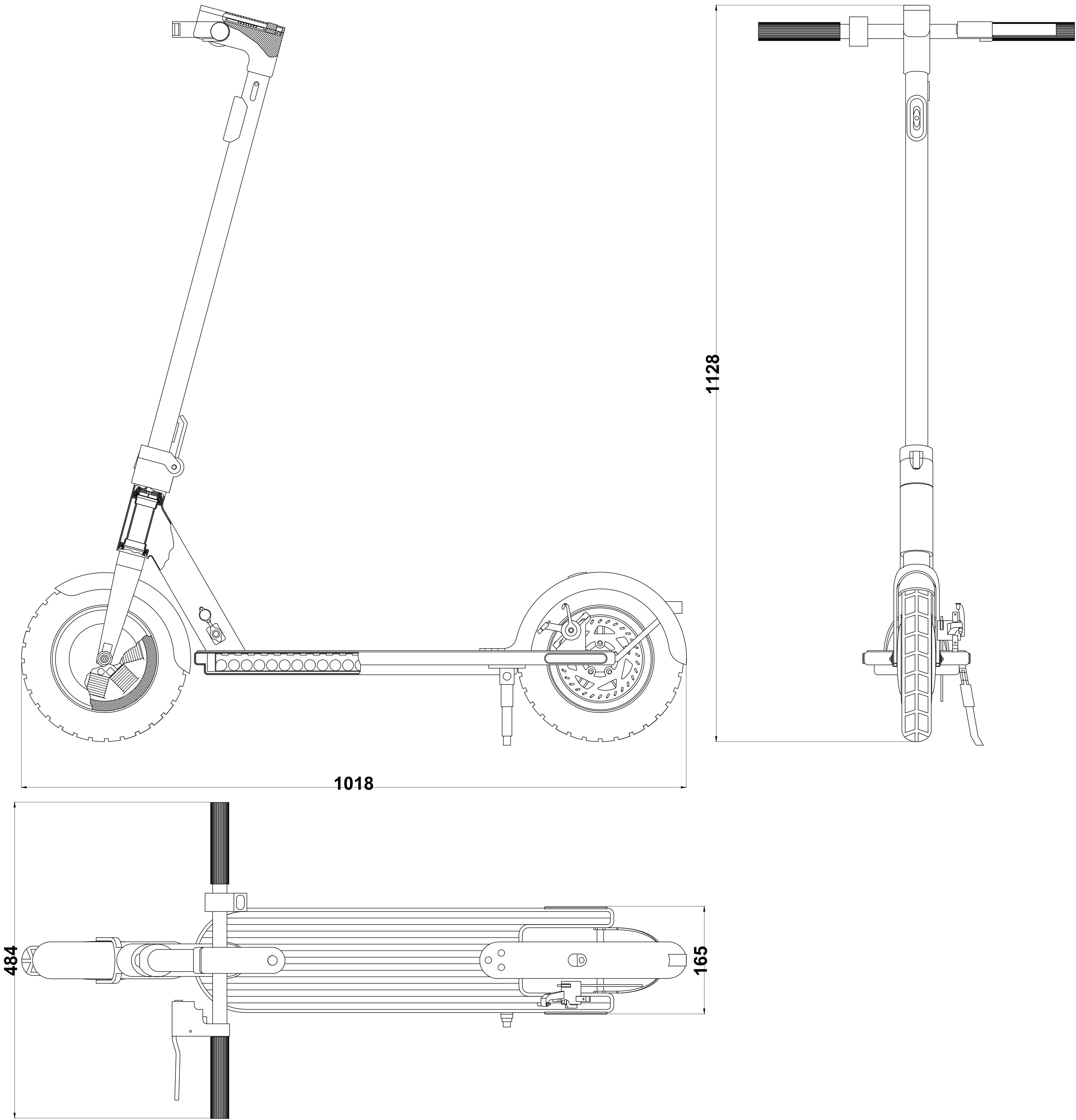


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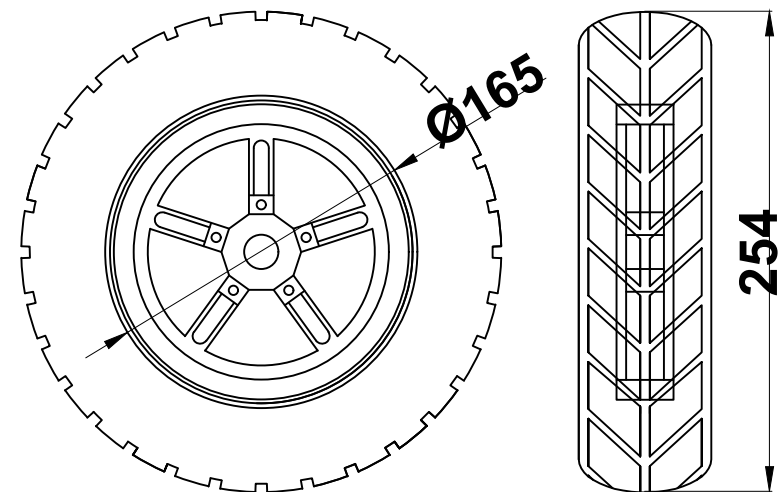
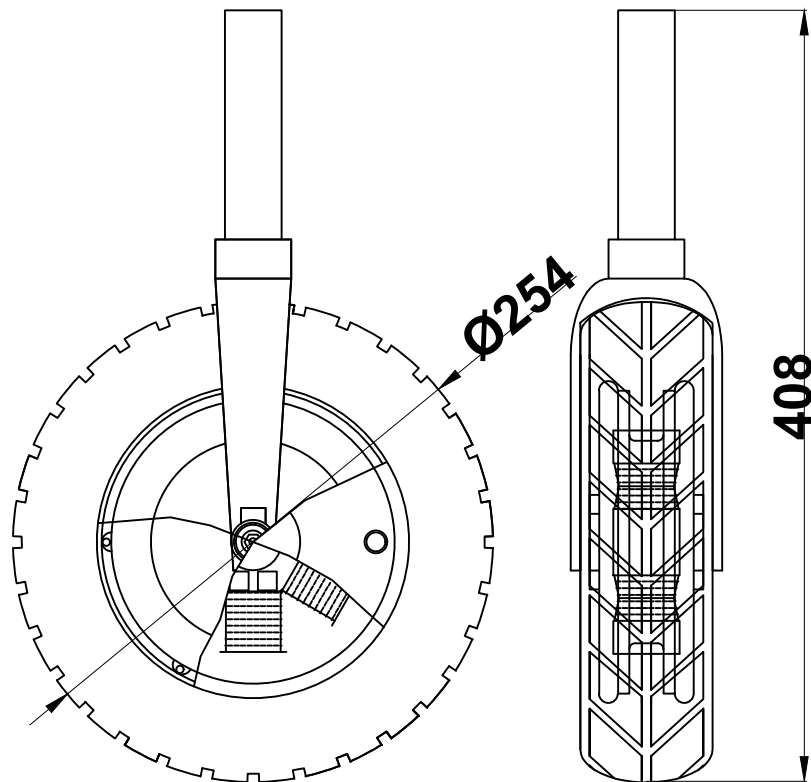
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| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                      | PL 21            | 3.6               | MM                                                                                    |
| PALANCA BLOQUEO                            |                  |                   | ESCALA 2:1                                                                            |
| INGENIERO/A RESPONSABLE: Hugo Ojeda Viedma |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                   | ACERO            | UNE 1-032-82      |                                                                                       |



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|--------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                      | PL 21            | 3.7               | MM                                                                                    |
| PASADOR Y EJE TRANSVERSAL                  |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Hugo Ojeda Viedma |                  |                   |  |
| JEFE AL CARGO: Hugo Ojeda Viedma           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                   | ACERO            | UNE 1-032-82      |                                                                                       |

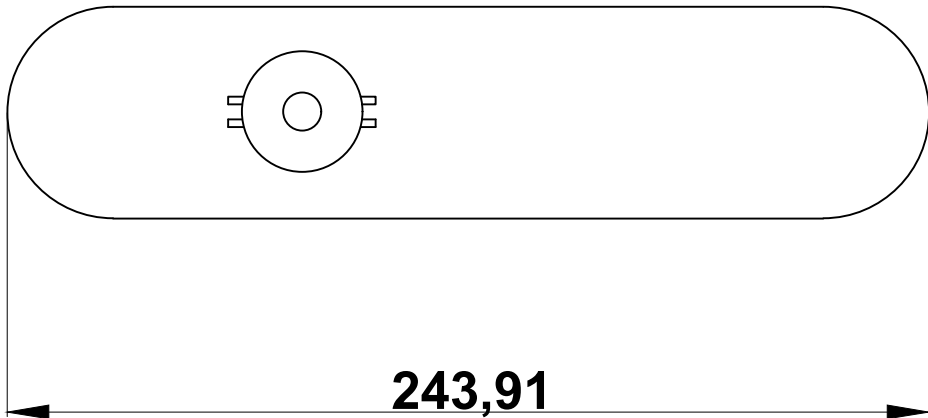
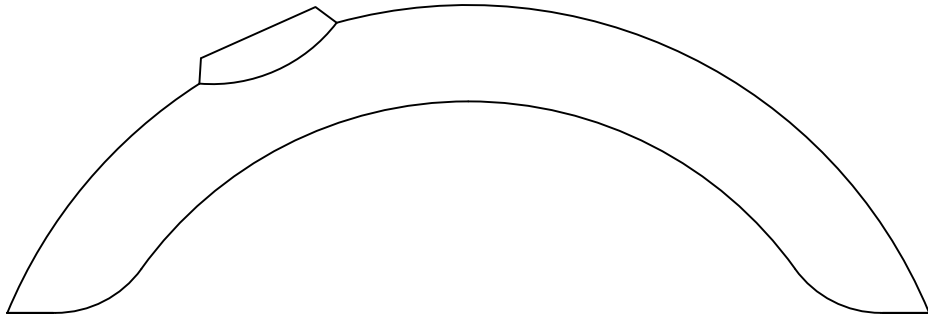
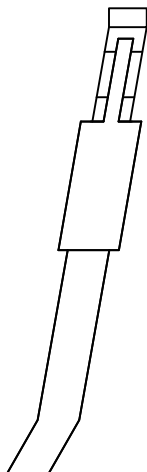
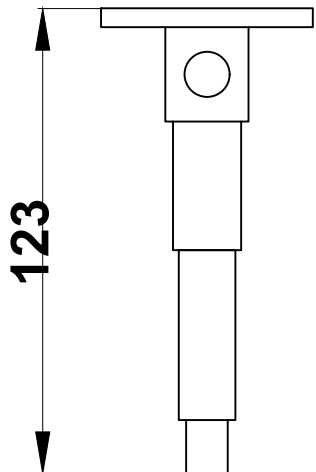
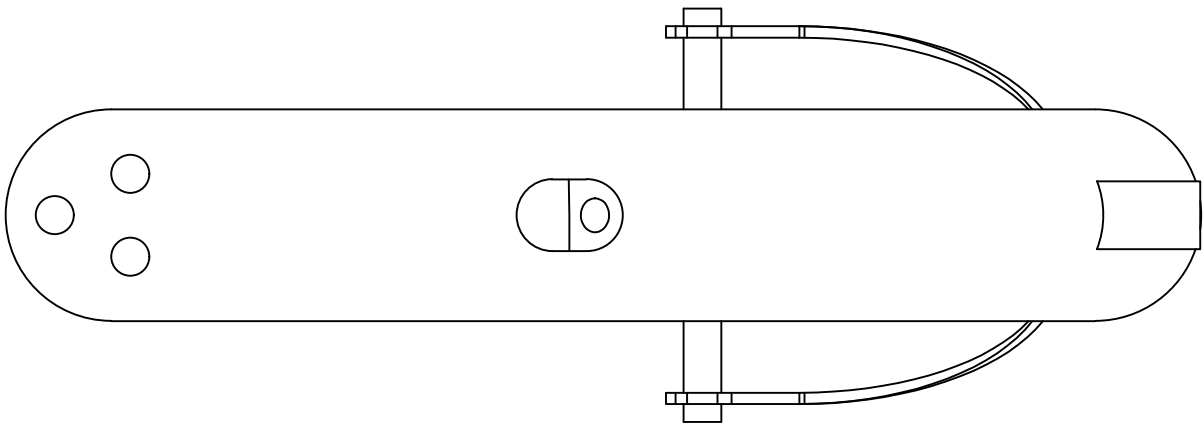
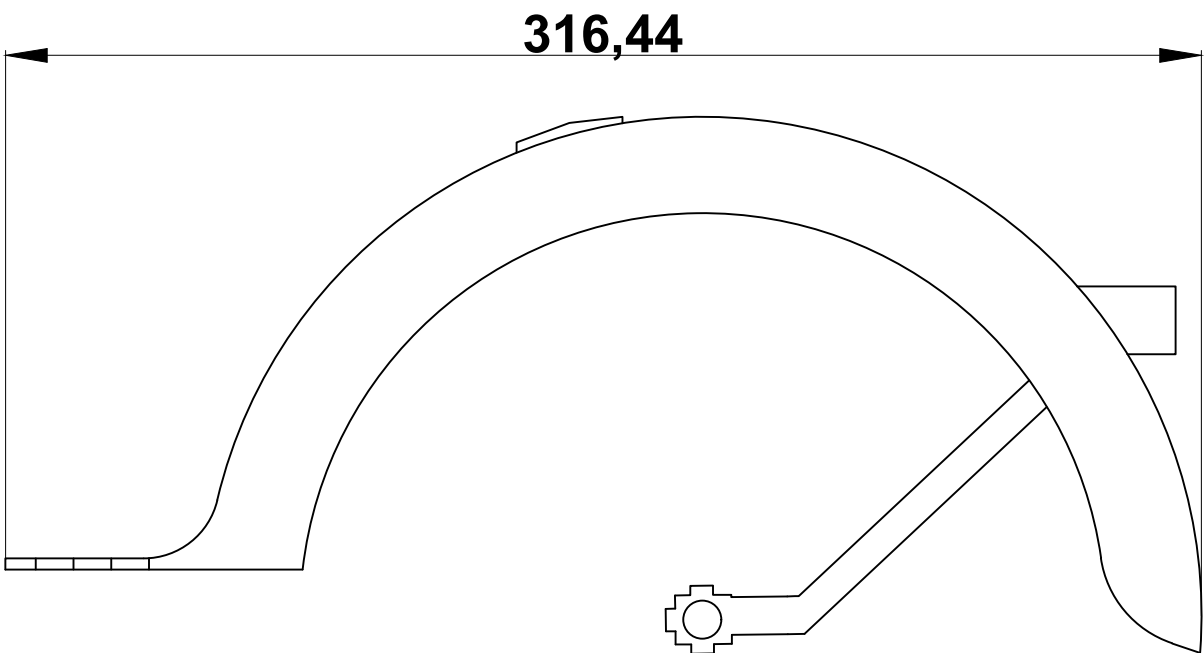


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|----------------------------------------------------|------------------|-------------------|------------|
| GRUPO PA                                           | GRUPO PL         | ID DEL DIBUJO     | UNIDADES   |
| PA 07                                              | PL 19, 20 y 21   | C1                | MM         |
| PATINETE COMPLETO                                  |                  |                   | ESCALA 1:3 |
| INGENIERO/A RESPONSABLE: Eloy Santos de Tourdonnet |                  |                   |            |
| DIRECTOR AL CARGO: Eloy Santos de Tourdonnet       |                  |                   |            |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |            |
| ALUMINIO                                           | ACERO            | UNE 1-032-82      |            |

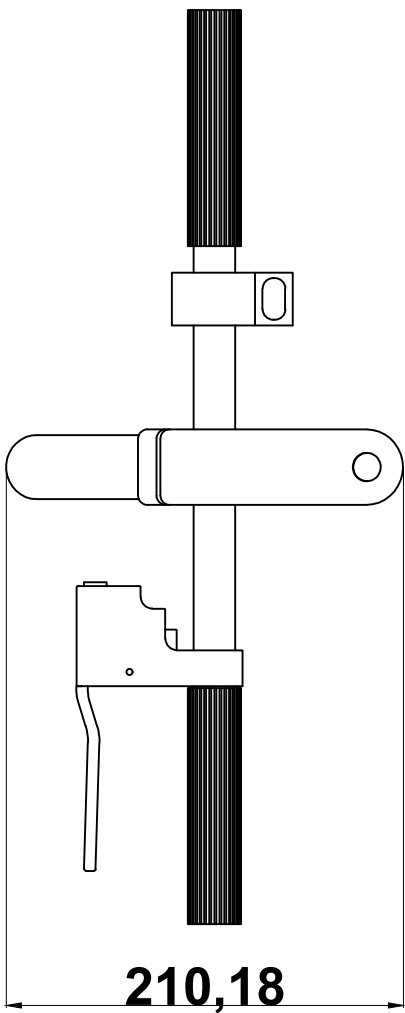
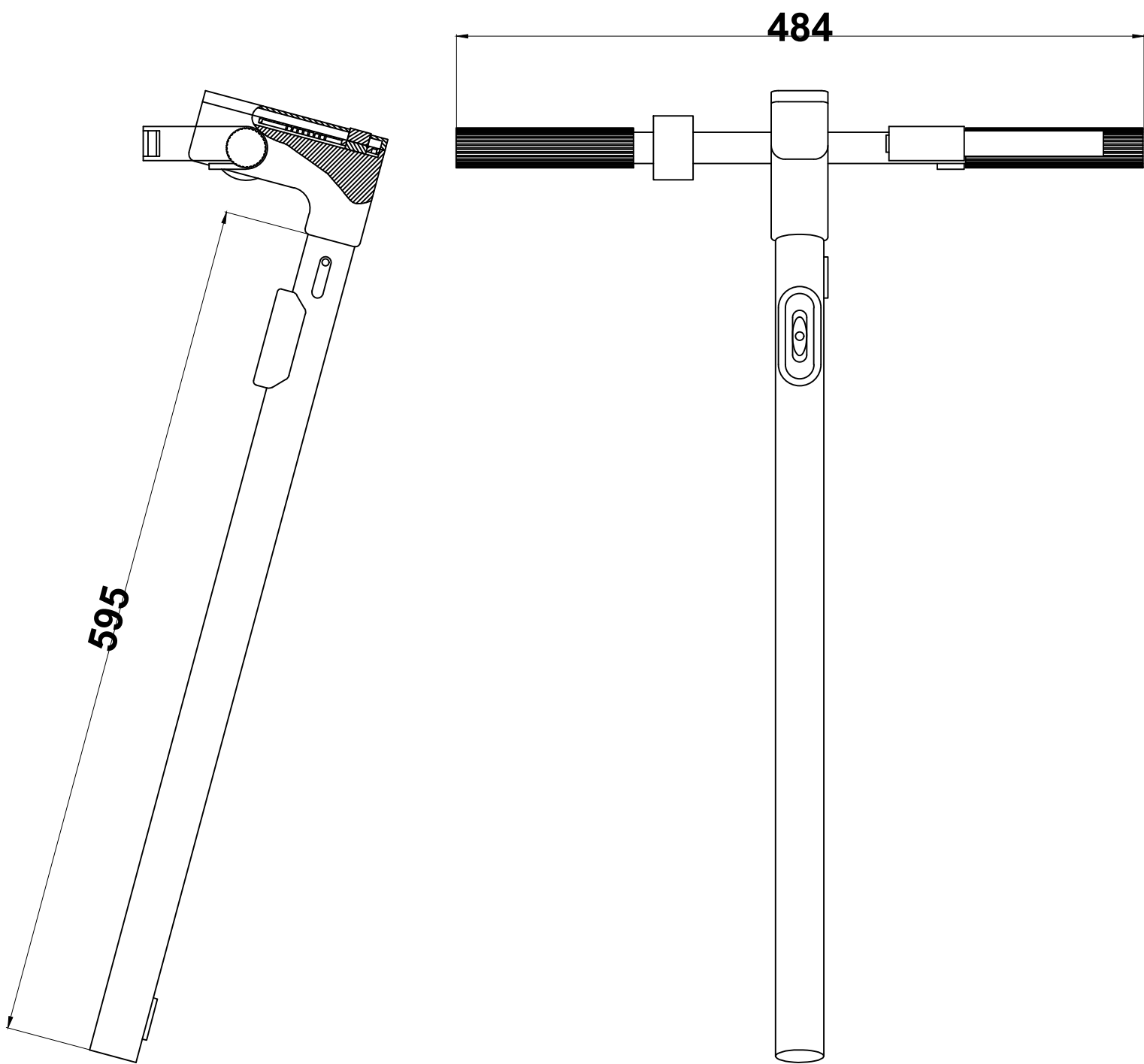


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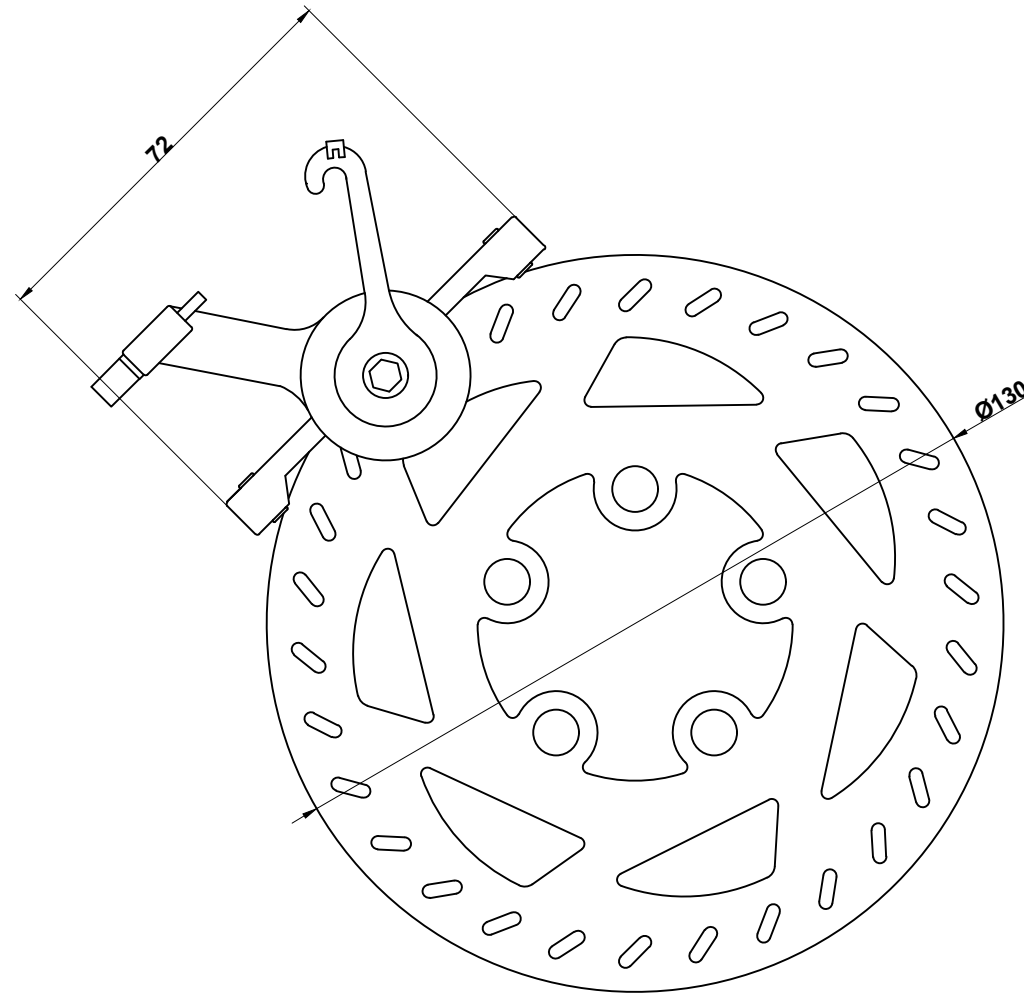
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| GRUPO PA                                        | GRUPOPL          | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                           | PL 19            | S1                | MM                                                                                    |
| TREN DE RODAJE                                  |                  |                   | ESCALA: 1:2                                                                           |
| INGENIERO/A RESPONSABLE: Marcos Menéndez Prieto |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez Prieto           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                              | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ACERO                                           | GOMA             | UNE 1-032-82      |                                                                                       |



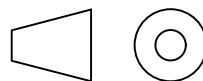
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| GRUPO PA                                        | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                           | PL 19            | S2                | MM                                                                                    |
| CARENADO Y ACCESORIOS                           |                  |                   | ESCALA 1:2                                                                            |
| INGENIERO/A RESPONSABLE: Marcos Menéndez Prieto |                  |                   |  |
| JEFE AL CARGO: Marcos Menéndez Prieto           |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                              | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| PLÁSTICO                                        | ACERO            | UNE 1-032-82      |                                                                                       |

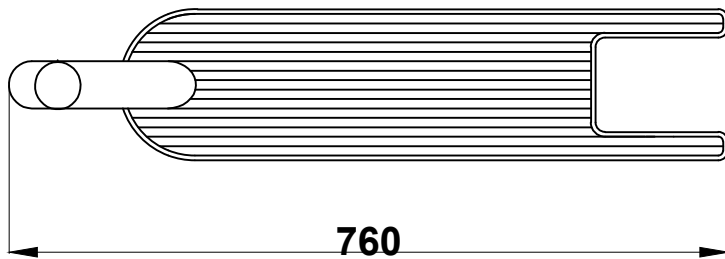
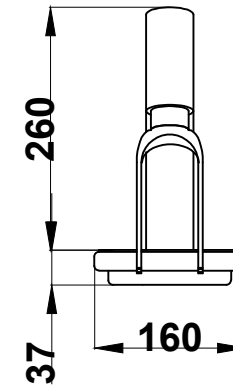
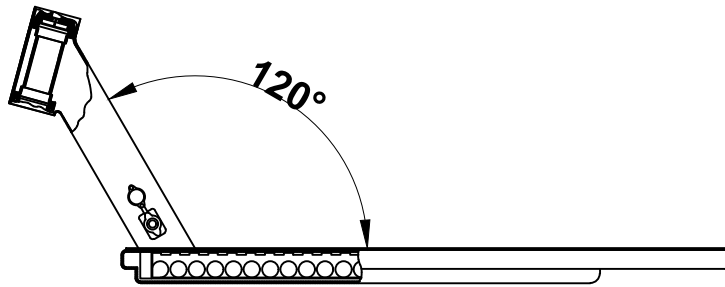


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|----------------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                           | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                              | PL 20            | S3                | MM                                                                                    |
| MANILLAR Y DIRECCIÓN                               |                  |                   | ESCALA 1:2                                                                            |
| INGENIERO/A RESPONSABLE: Eloy Santos de Tourdonnet |                  |                   |  |
| DIRECTOR AL CARGO: Eloy Santos de Tourdonnet       |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                           | ACERO            | UNE 1-032-82      |                                                                                       |



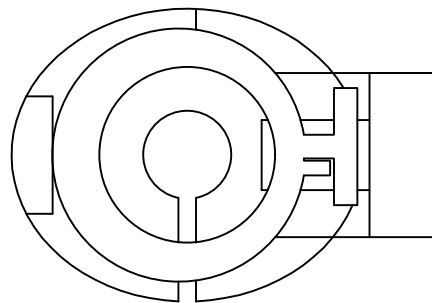
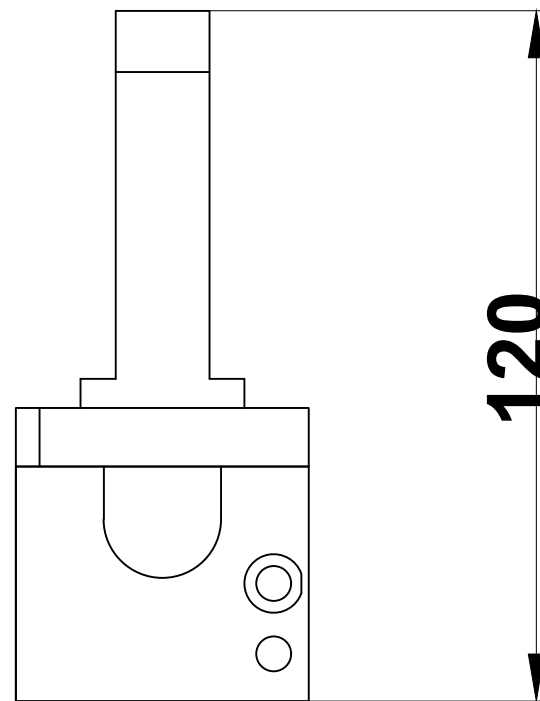
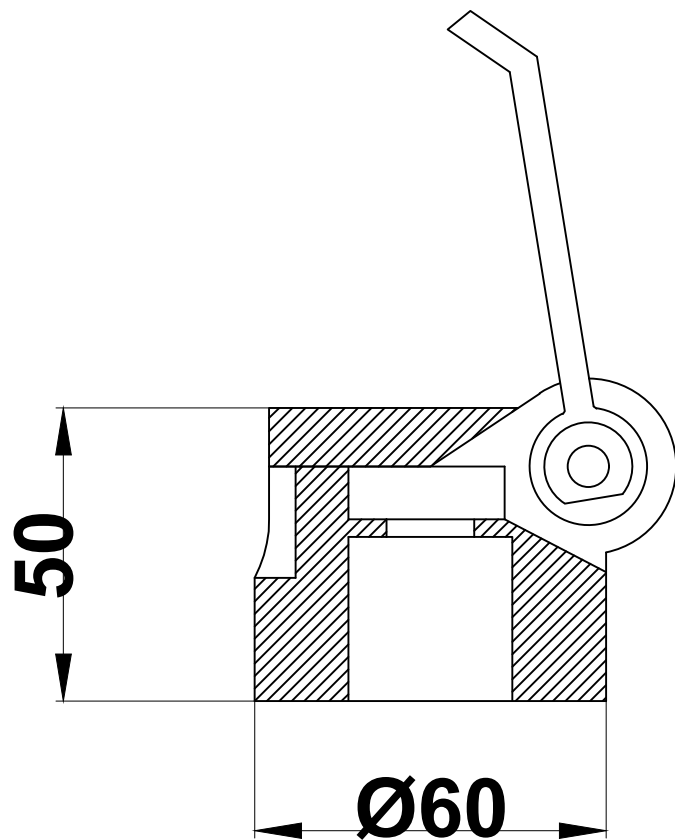
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|----------------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                           | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                              | PL 20            | S4                | MM                                                                                    |
| SISTEMA DE FRENADO                                 |                  |                   | ESCALA 3:4                                                                            |
| INGENIERO/A RESPONSABLE: Eloy Santos de Tourdonnet |                  |                   |  |
| DIRECTOR AL CARGO: Eloy Santos de Tourdonnet       |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                                 | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                           | ACERO            | UNE 1-032-82      |                                                                                       |

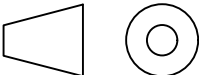


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| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
|--------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| PA 07                                      | PL 21            | S5                | MM                                                                                    |
| Chasis y Cuello                            |                  |                   | ESCALA 1:8                                                                            |
| INGENIERO/A RESPONSABLE: Hugo Ojeda Viedma |                  |                   |  |
| DIRECTOR AL CARGO: Hugo Ojeda Viedma       |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                   | ACERO            | UNE 1-032-82      |                                                                                       |



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|--------------------------------------------|------------------|-------------------|---------------------------------------------------------------------------------------|
| GRUPO PA                                   | GRUPO PL         | ID DEL DIBUJO     | UNIDADES                                                                              |
| PA 07                                      | PL 21            | S6                | MM                                                                                    |
| MECANISMO DE PLEGADO                       |                  |                   | ESCALA 1:1                                                                            |
| INGENIERO/A RESPONSABLE: Hugo Ojeda Viedma |                  |                   |  |
| DIRECTOR AL CARGO: Hugo Ojeda Viedma       |                  |                   |                                                                                       |
| MATERIAL PRINCIPAL                         | OTROS MATERIALES | NORMATIVA SEGUIDA |                                                                                       |
| ALUMINIO                                   | ACERO            | UNE 1-032-82      |                                                                                       |